

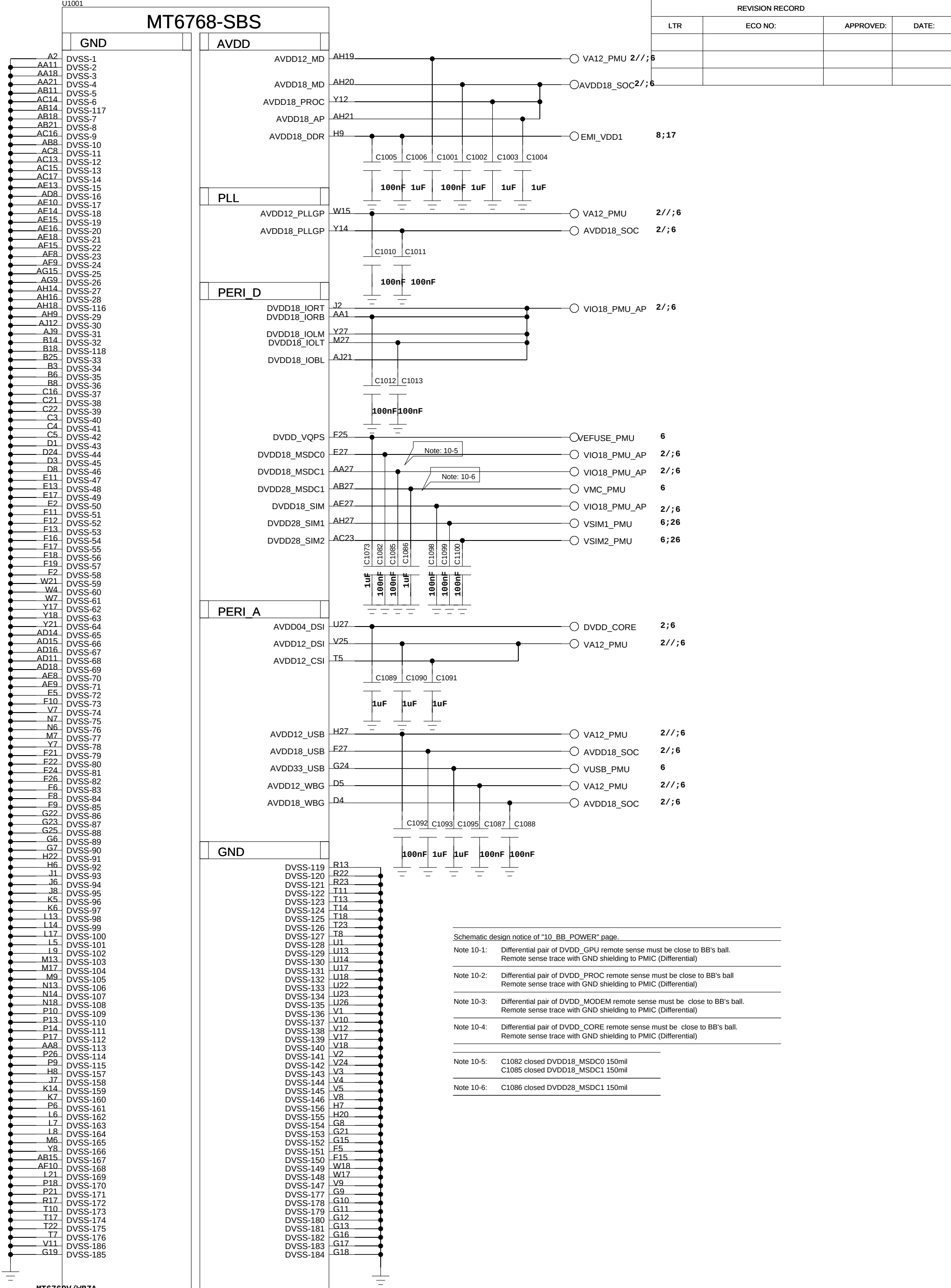
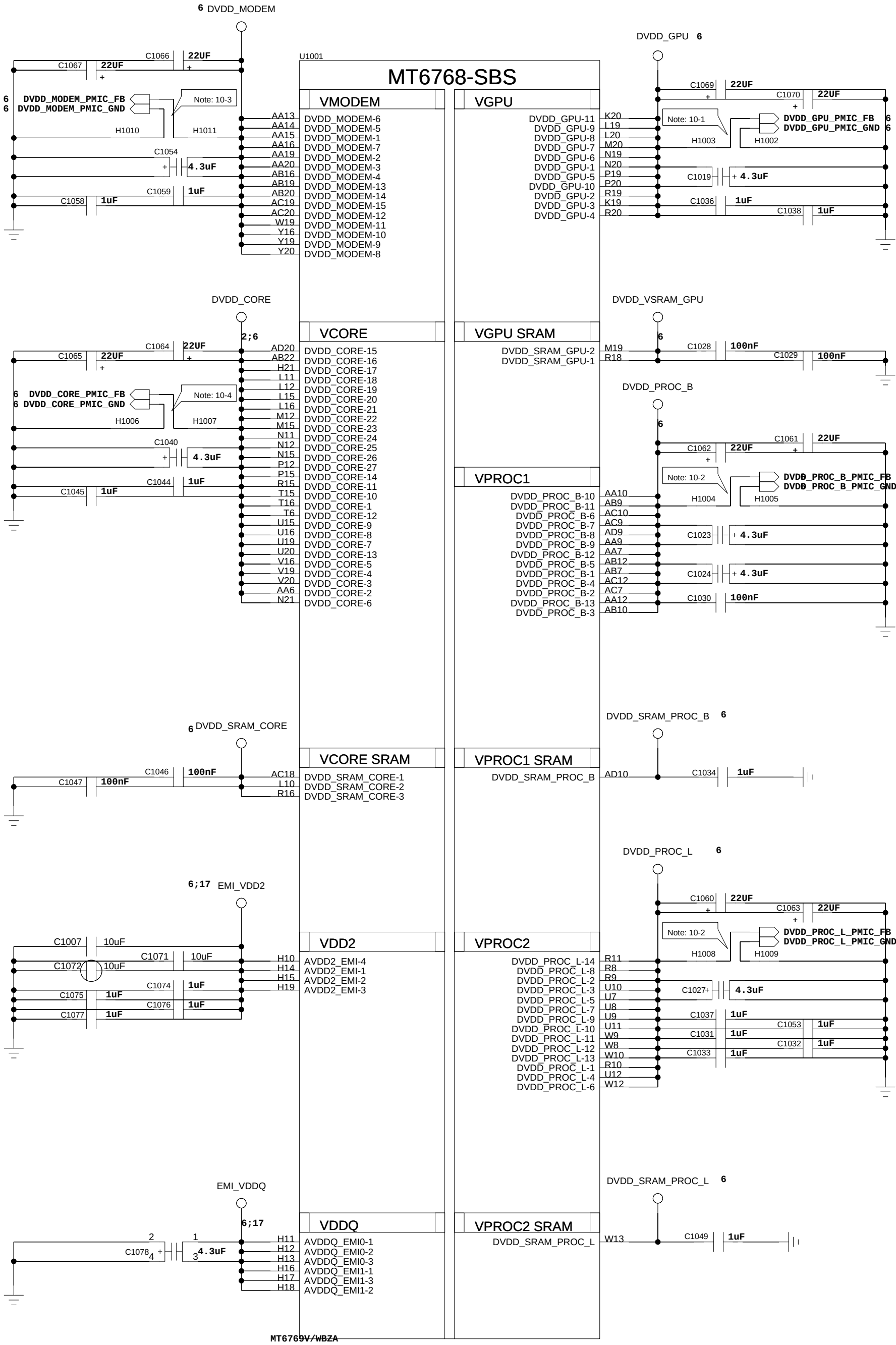
Project : MT6768/69 LPDDR4x

The diagram illustrates the system architecture of the MT6768/69 (SBS) processor. It is organized into three main functional blocks:

- Peripheral Components (Left):** This section details the various external components connected to the processor. Key components include:
 - Connectivity:** Connectivity ANT, TRIPLEXER, SAW, LNA, 2.4G, 5G, FM, and Headset.
 - Display & Camera:** LCD/AMOLED+CTP, REAR CAM I, REAR CAM II, FRONT CAM, and FLASH.
 - Storage & Memory:** SIM1, SIM2, Micro SD, and eMCP LPDDR4x+eMMC.
 - Sensors & Input:** ALS/PS, E-Compass, Gsensor, CHARGER, LCM Gate Driver, BACK/FRONT FLASH, Smart PA, FINGER PRINT, Keypad, and Debug port.
- Core Processor (Center):** The central component is the MT6768/69 (SBS) processor, which is connected to various internal modules.
- System Integrators/Modules (Right):** This section shows the integration of various modules and components:
 - RF & Antennas:** RF IQ, BSI ctrl, MT6177M (LWG+LTG), CLK Ctrl, RXD, RX, TX, RxD FEM, FEM, RXD ANT, and Main 1 ANT.
 - Power & Clock:** 26M_RF, 26M_BB, 26M_WBG, 32K_BB, DCXO / TMS, 26MHz Crystal, LDOs, RTC, Vibrator, VIB, Headset, Receiver, AU_VIN0, VSYS, and Battery.
 - Audio & I/O:** Headset (HPL, HPR, AU_VIN1), Receiver, AU_VIN0, and USB 2.0 / USB3.0.
 - Other Components:** LCM Gate Driver, Flash, WLED BST, VSYS, Charger/SW, and USB Conn.

COMPANY: TRANSSION HOLDINGS				MODEL: G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 00_BLOCK_DIAGRAM		VERSION:	SHEET: 1 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

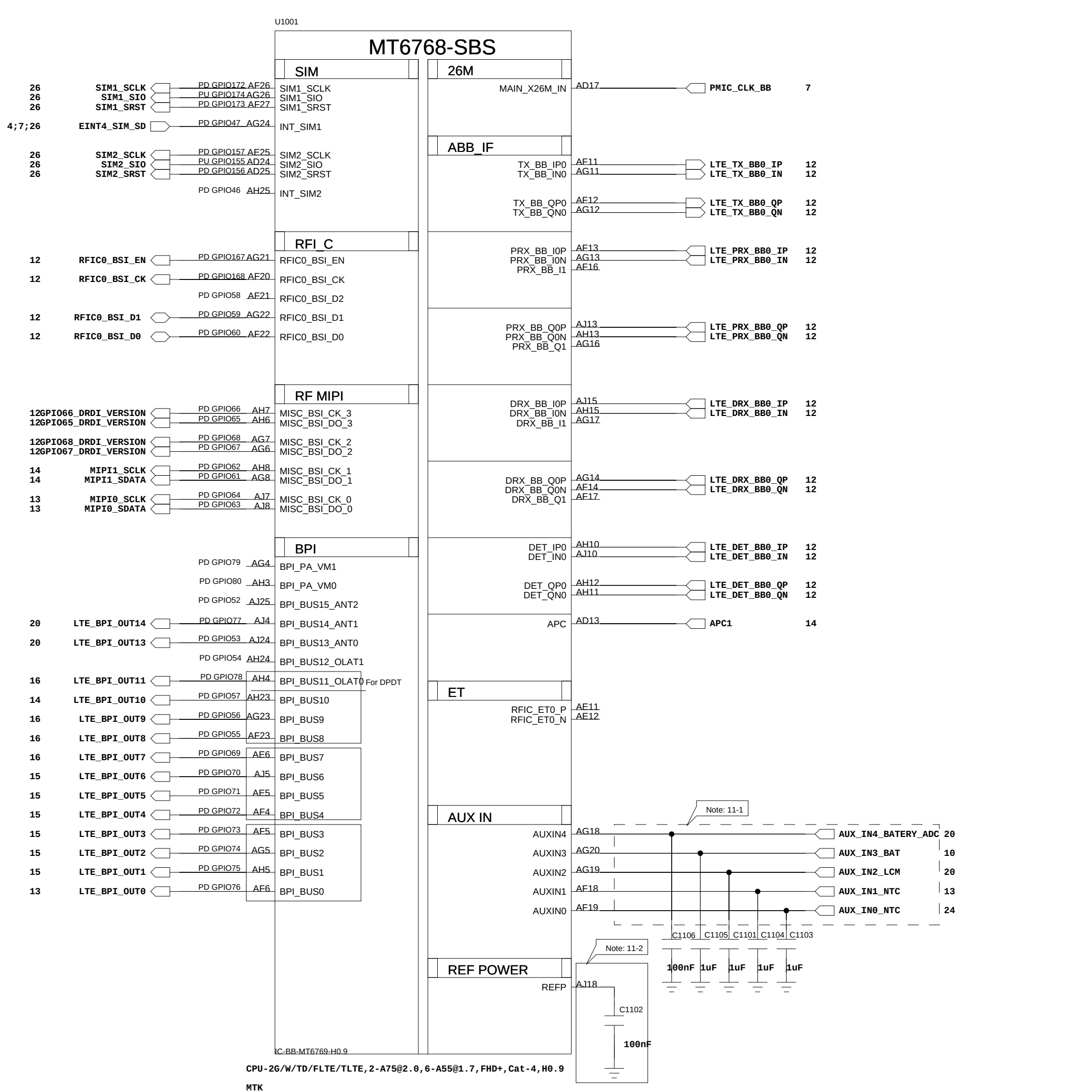
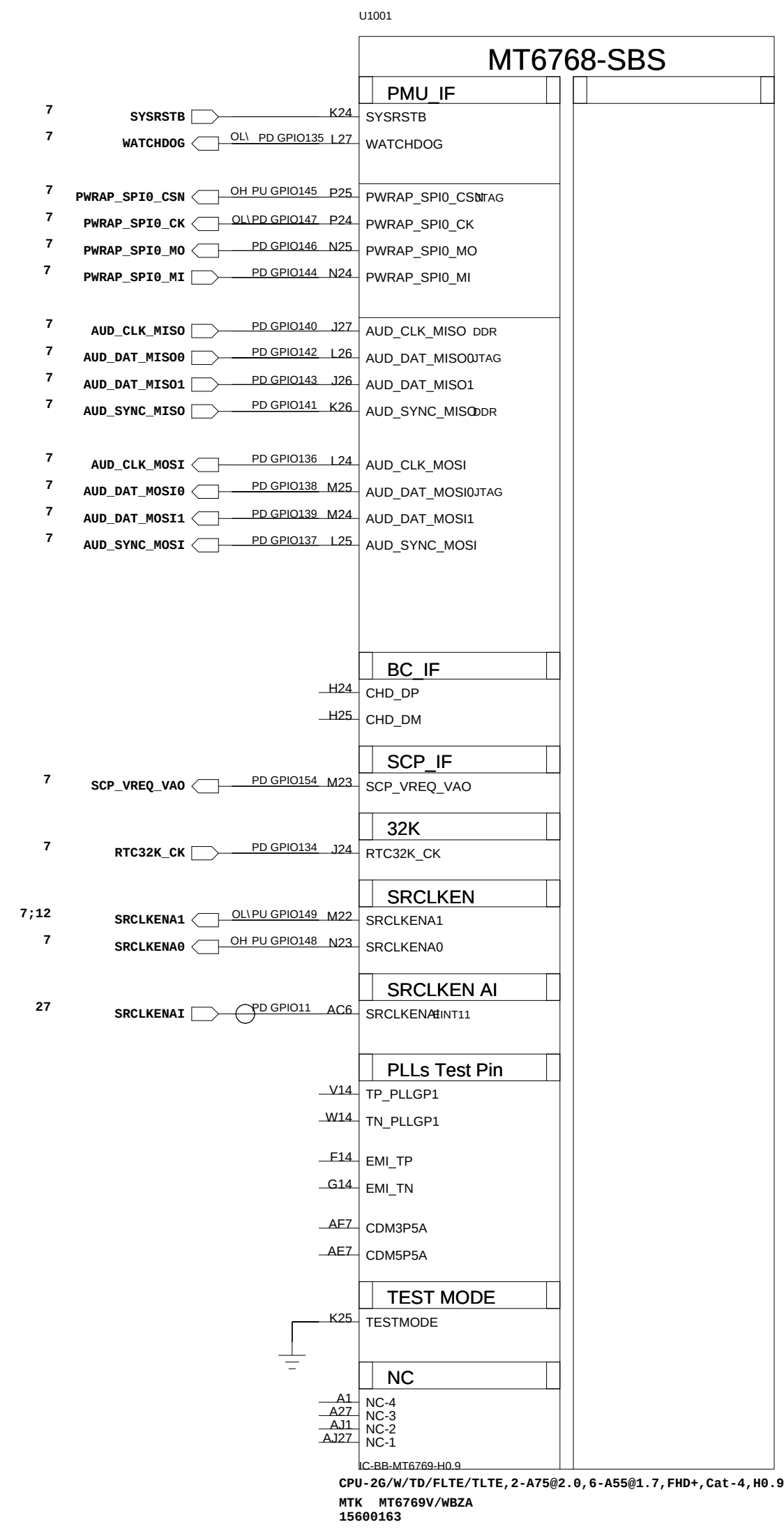
BB_POWER



REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 10_BB_POWER		VERSION:	SHEET: 2 OF 27
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REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



"PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pin to select which interface will be the JTAG pin out.			
	PWRAP_SPI0_CSN	AUD_DAT_MOSI0	AP_JTAG
Default	HI	LO	N/A
	HI	HI	SPI0+EINT8
	LO (No test)	LO (No test)	SPI0+EINT8
	LO (No test)	HI (No test)	N/A

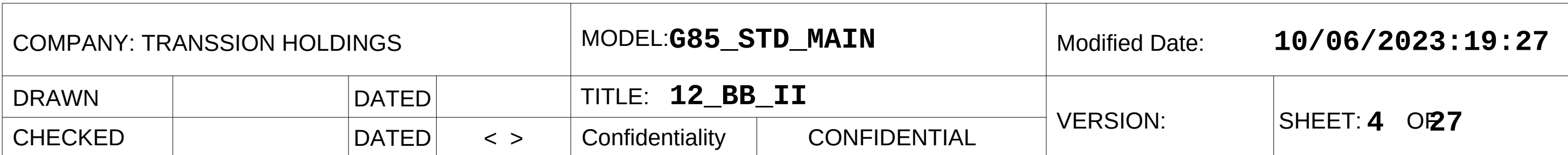
AUD_SYNC_MISO and AUD_CLK_MISO are DDR type feature in bootstrap			
	AUD_SYNC_MISO	AUD_CLK_MISO	DDR Type
Default	LO	LO	LP4X eMCP
	LO	HI (No test)	Reserved
	HI (No test)	LO	LP3 eMCP
	HI (No test)	HI (No test)	Reserved

Schematic design notice of "11_BB_I" page.

Note 11-1: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 11-2: The de-coupling cap. for REFP (AJ21 ball) have to be placed as close to BB as possible.

COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 11_BB_I		VERSION:	SHEET: 3 OF 27
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U1001

MT6768-SBS

EMI IF		EMI IF	
17	EMI1_DQ15	EMI1_CS1	C20
17	EMI1_DQ14	EMI1_CS0	D19
17	EMI1_DQ13	EMIO_CS1	C12
17	EMI1_DQ12	EMIO_CS0	F9
17	EMI1_DQ11		
17	EMI1_DQ10		
17	EMI1_DQ9	EMI1_CKE1	D20
17	EMI1_DQ8	EMI1_CKE0	F20
17	EMI1_DQ7		
17	EMI1_DQ6	EMIO_CKE1	D9
17	EMI1_DQ5	EMIO_CKE0	C9
17	EMI1_DQ4		
17	EMI1_DQ3	EMI1_DMI1	C17
17	EMI1_DQ2	EMI1_DMI0	B22
17	EMI1_DQ1		
17	EMI1_DQ0	EMIO_DMI1	F12
		EMIO_DMI0	B10
17	EMI1_CA5	EMI1_DQS1_C	F15
17	EMI1_CA4	EMI1_DQS1_T	D15
17	EMI1_CA3	EMI1_DQS0_C	E23
17	EMI1_CA2	EMI1_DQS0_T	D23
17	EMI1_CA1		
17	EMI1_CA0	EMIO_DQS1_C	D14
		EMIO_DQS1_T	F14
17	EMIO_DQ15	EMIO_DQS0_C	D6
17	EMIO_DQ14	EMIO_DQS0_T	F6
17	EMIO_DQ13		
17	EMIO_DQ12	EMI1_CK_C	D18
17	EMIO_DQ11	EMI1_CK_T	F18
17	EMIO_DQ10		
17	EMIO_DQ9	EMIO_CK_C	D10
17	EMIO_DQ8	EMIO_CK_T	F10
17	EMIO_DQ7		
17	EMIO_DQ6		
17	EMIO_DQ5	EMI_RESET_N	C24
17	EMIO_DQ4		
17	EMIO_DQ3		
17	EMIO_DQ2		
17	EMIO_DQ1		
17	EMIO_DQ0		
17	EMI0_CA5		
17	EMI0_CA4		
17	EMI0_CA3		
17	EMI0_CA2		
17	EMI0_CA1		
17	EMI0_CA0		
EMI EXTR			
EMI_EXTR			

A6

The resistor of EMI_EXTR for DRAM has to be placed near to BB as close as possible, please select 60.4 ohm 1% resistor

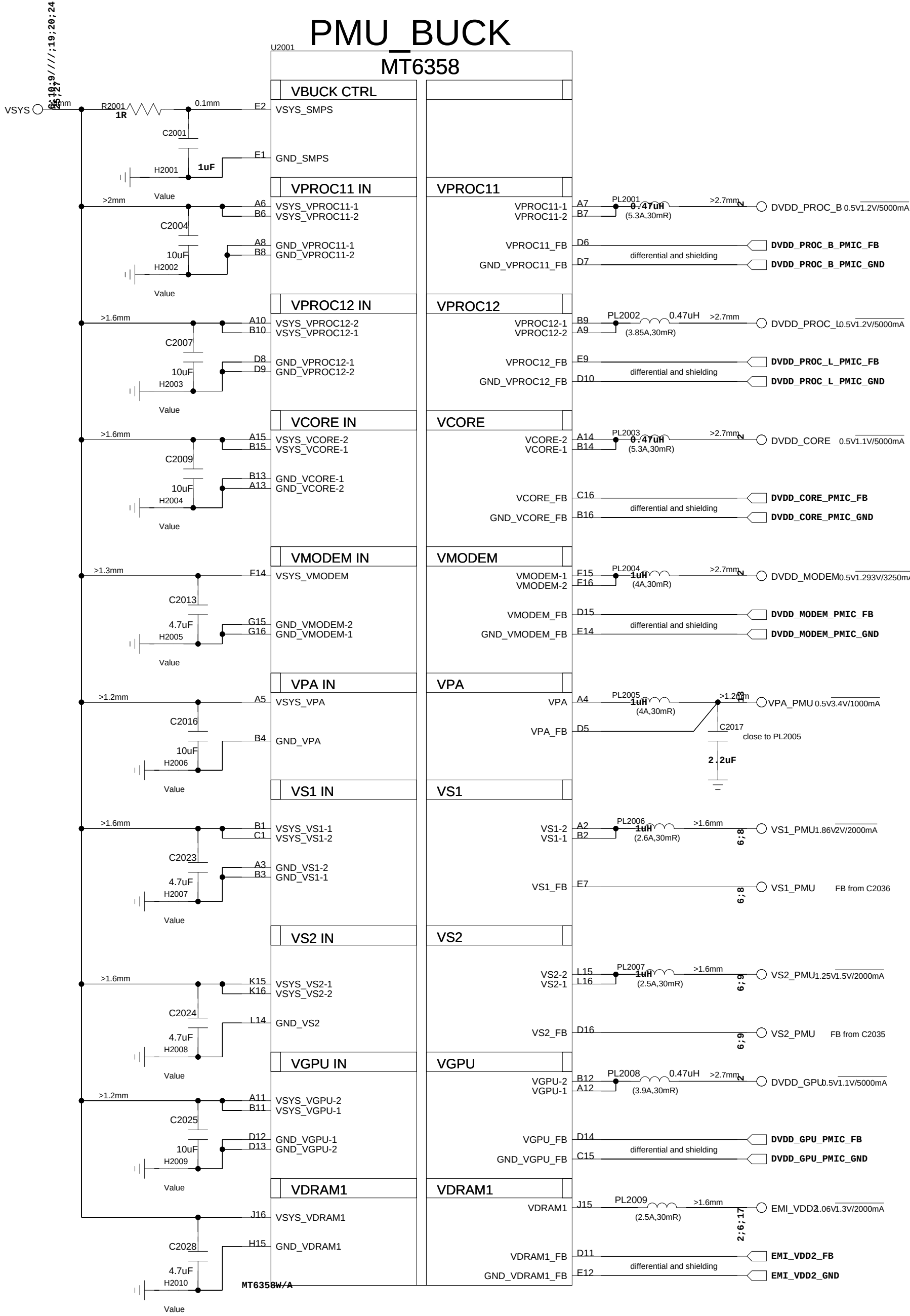
MT6769V/WBZA

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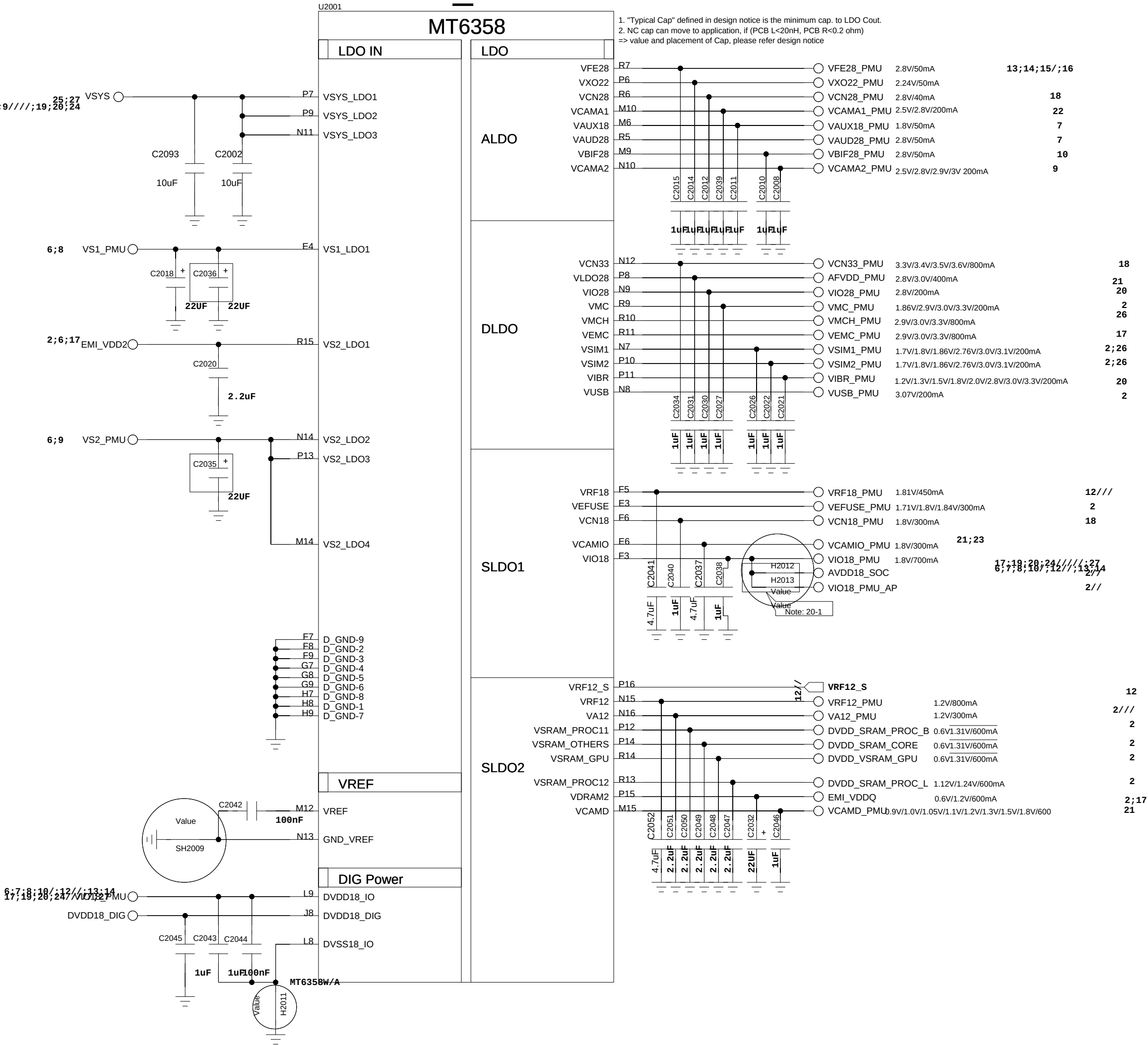
POWER_MT6358_I

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

PMU BUCK



PMU_LDO



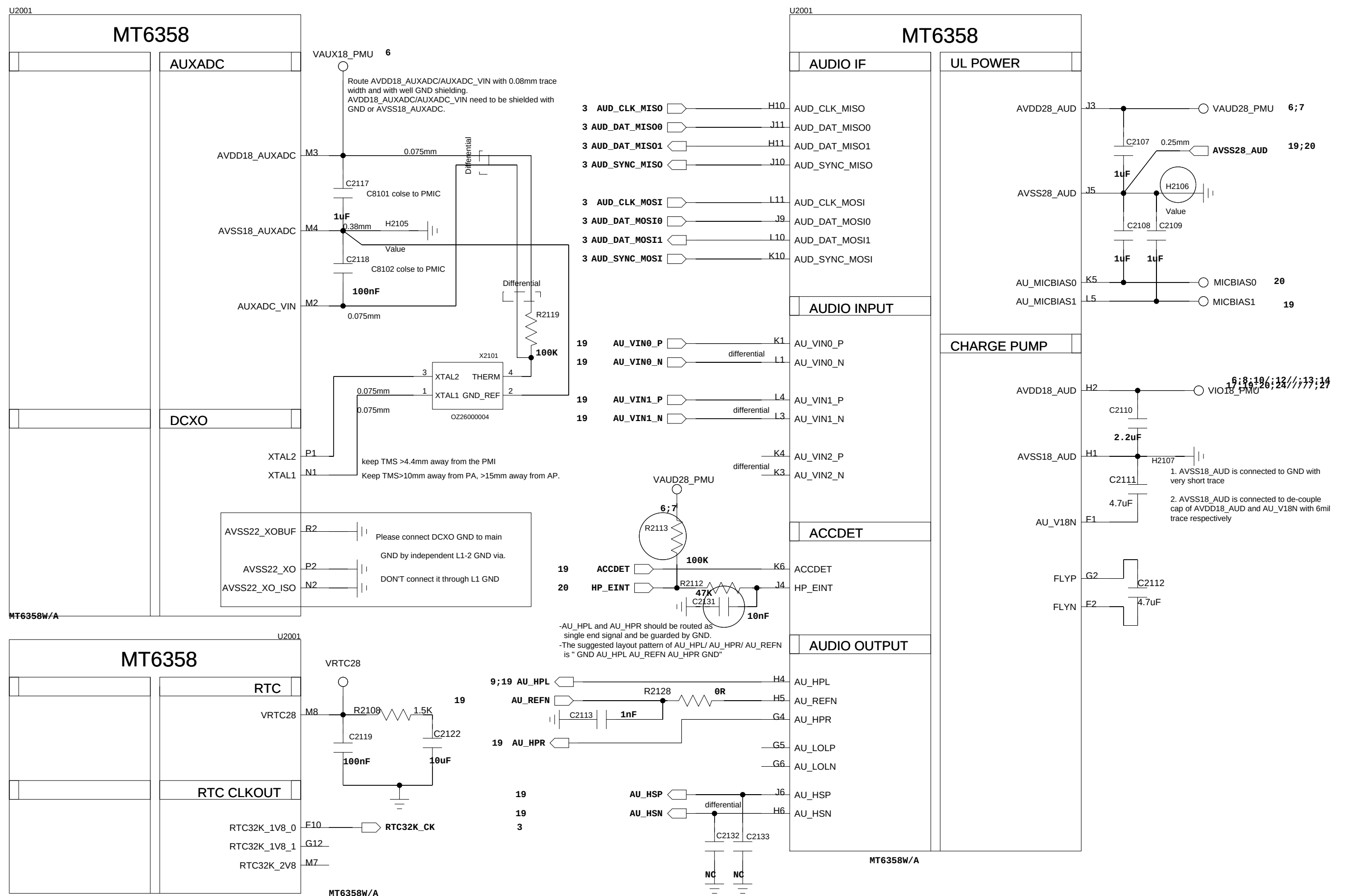
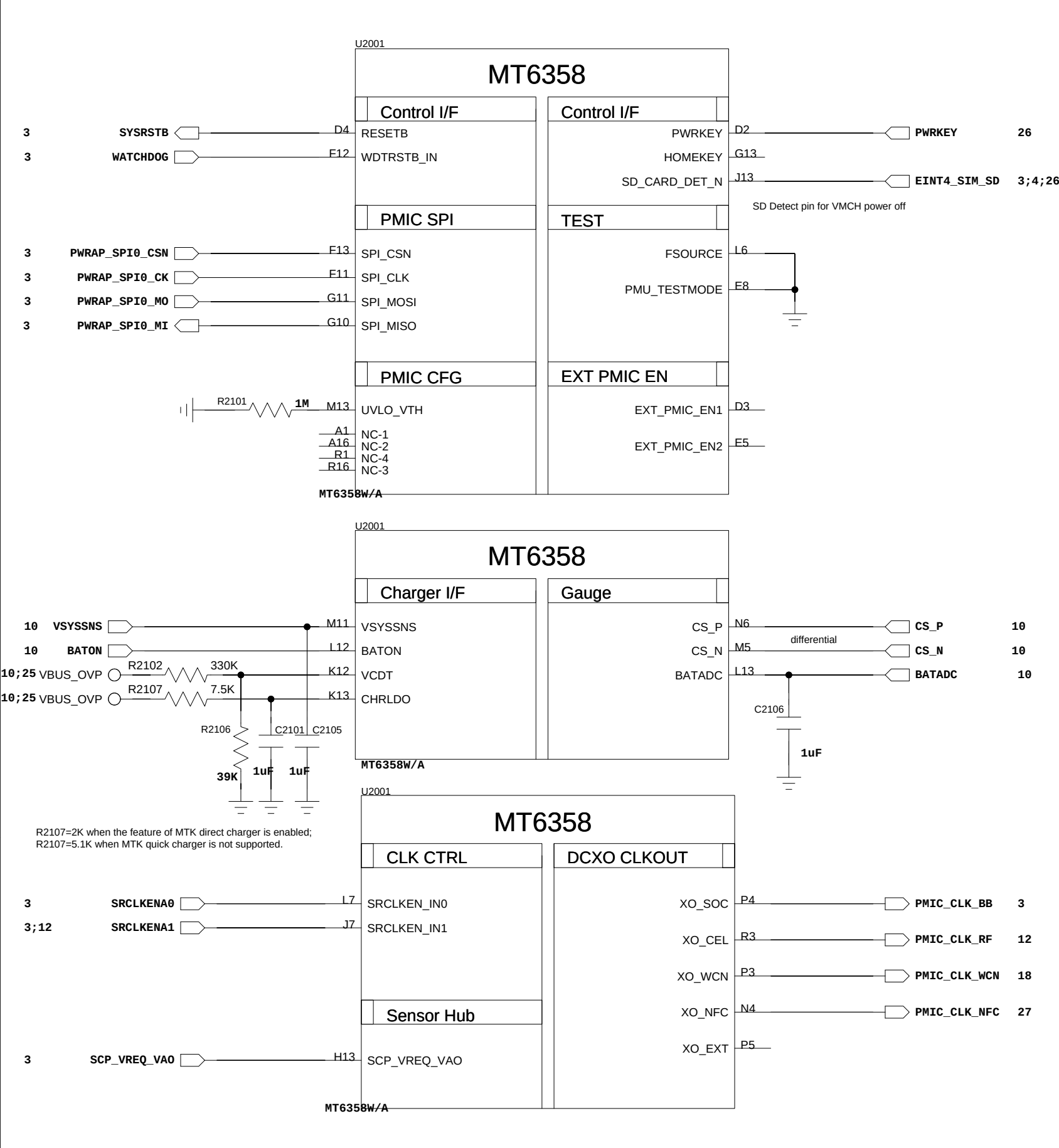
Schematic design notice of "21_POWER_MT6358-LDO" page.

Note 21-1: Please set H2012/H2013 close to C2038, making star connection between VIO18_PMU and AVDD18_SOC/VIO18_PMU_AP near to LDO cap. C2038
Please also refer to MT6358 design notice for further detail design information

COMPANY: TRANSSION HOLDINGS				MODEL: G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 20_POWER_MT6358_I		VERSION:	SHEET: 6 OF 27
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POWER_MT6358_II

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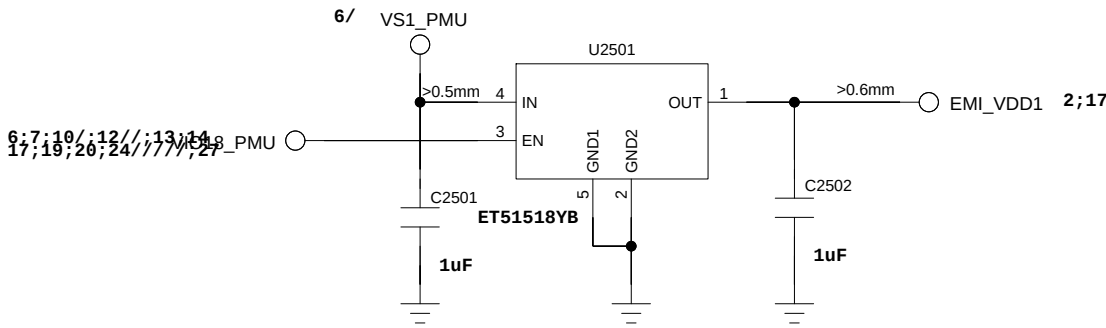


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POWER_THIRD-PARTY-I

System supply

LPDDR4X VDD1 1.8V LDO

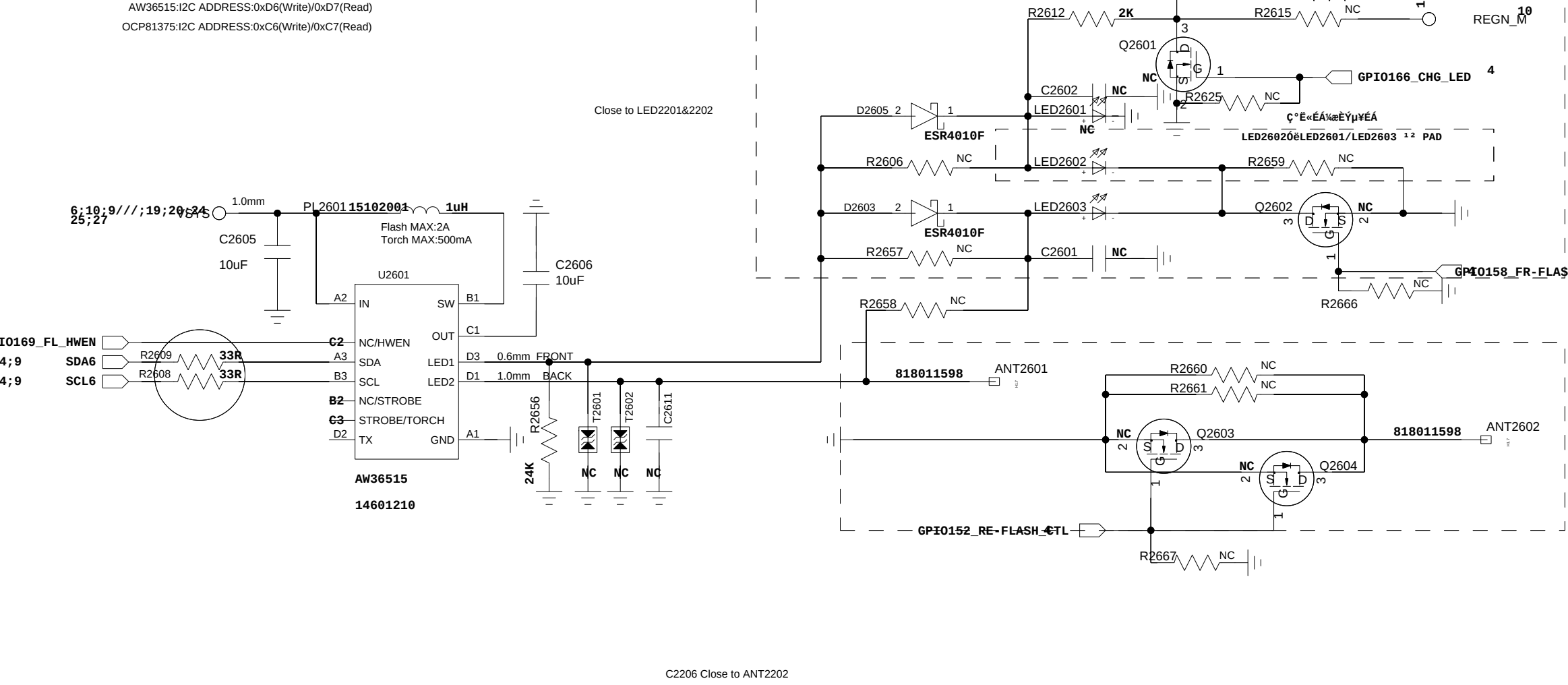


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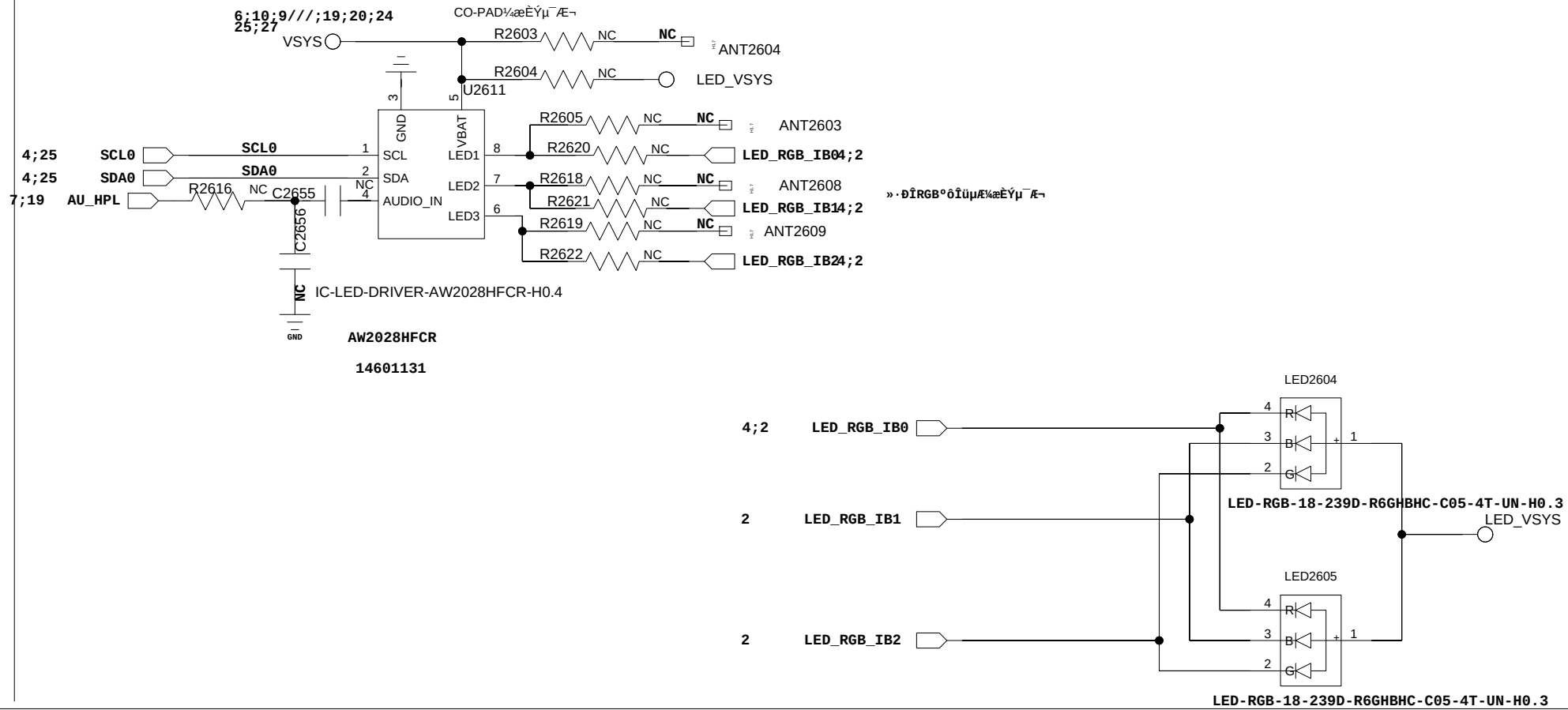
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CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

POWER_THIRD-PARTY_I

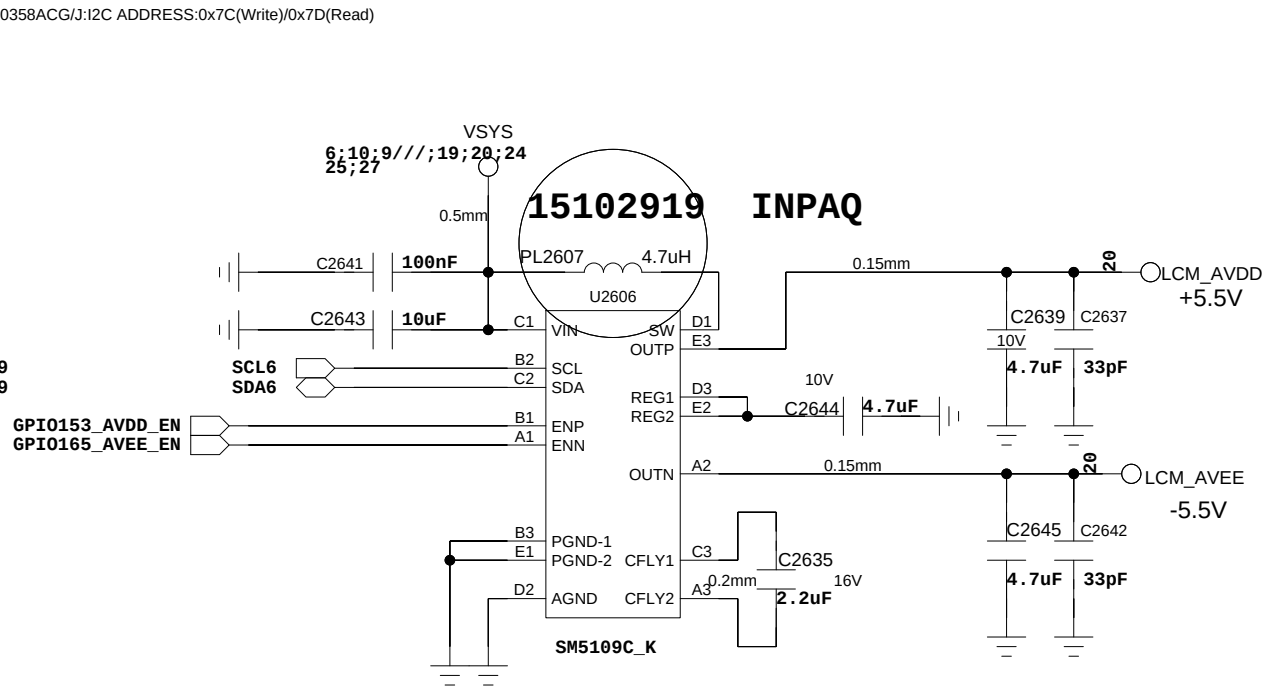
FRONT CAMERA FLASH LED



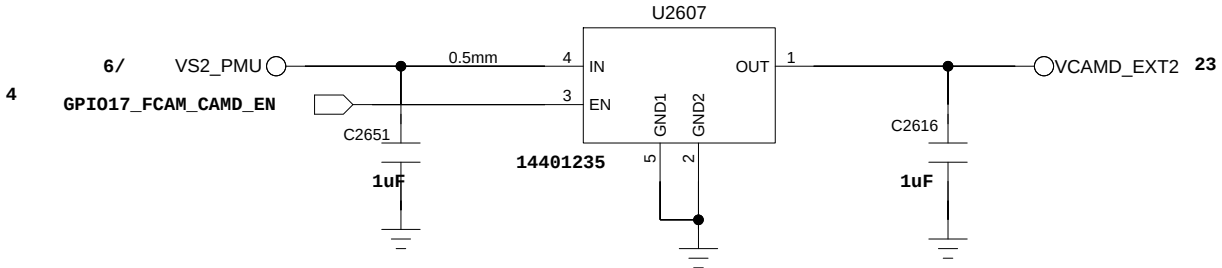
RGB LED



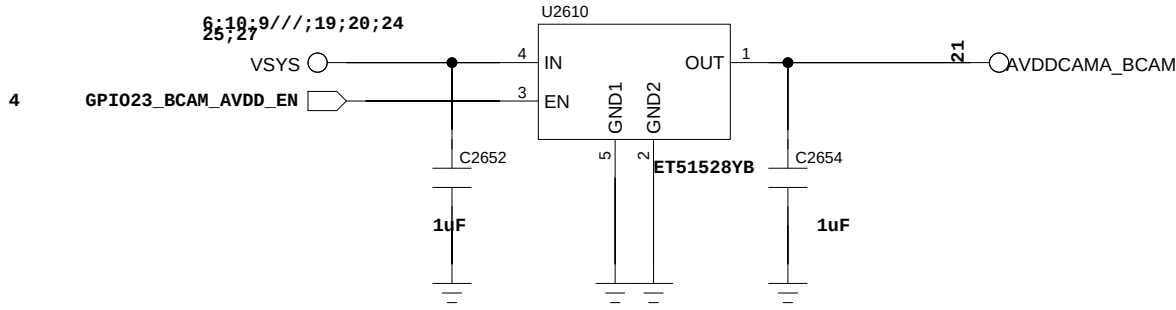
LCM BIAS



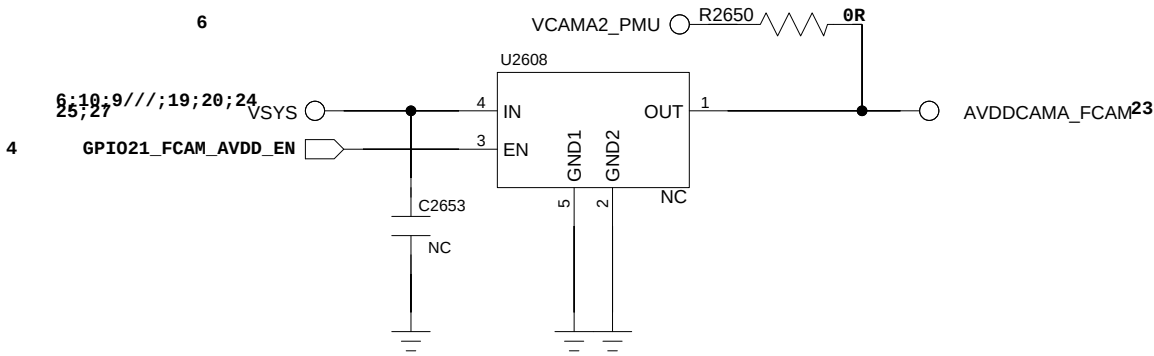
LDO FCAM DVDD(32M)



LDO BCAM AVDD

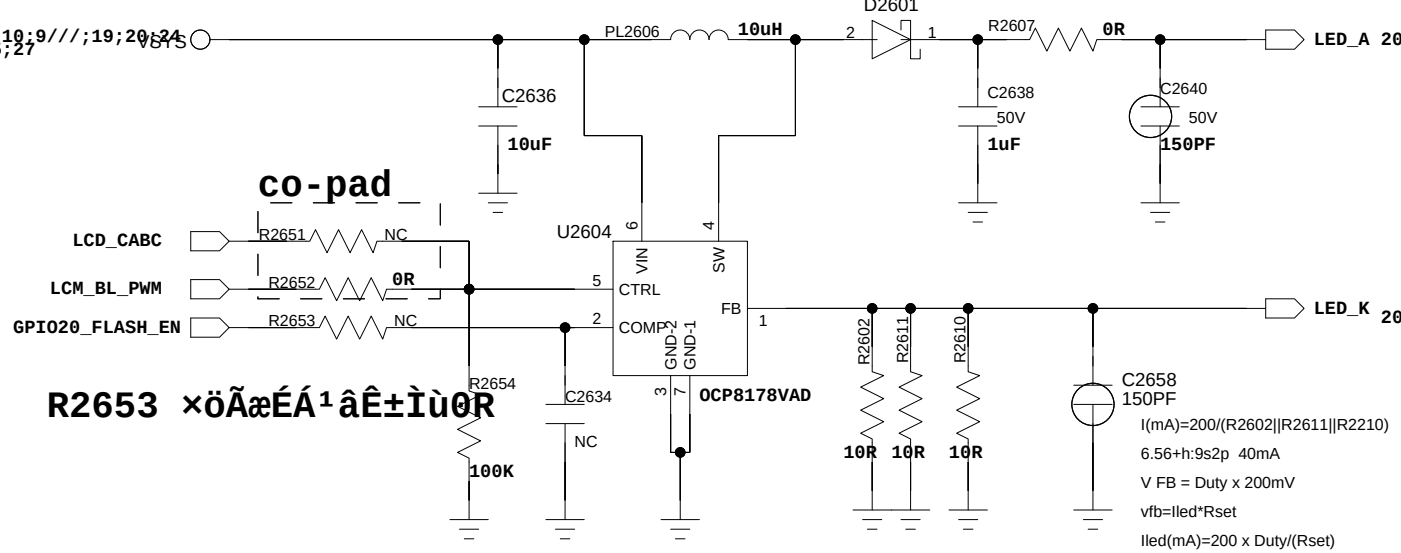


LDO FCAM AVDD



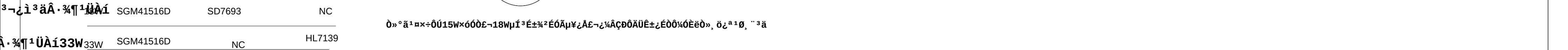
LDO

LCM LED



COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
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LTR	ECO NO:	APPROVED:	DATE:

[illegible]

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CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

POWER_THIRD-PARTY-IV

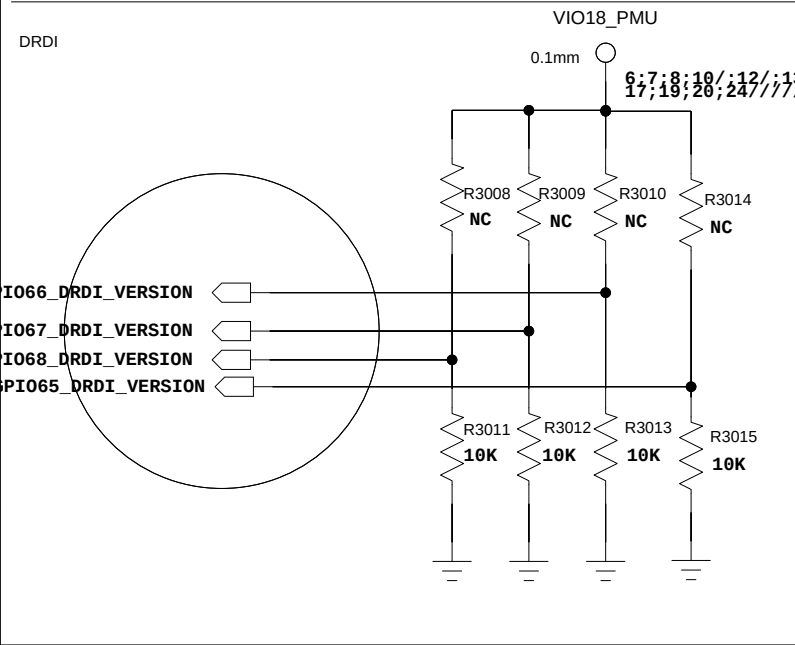
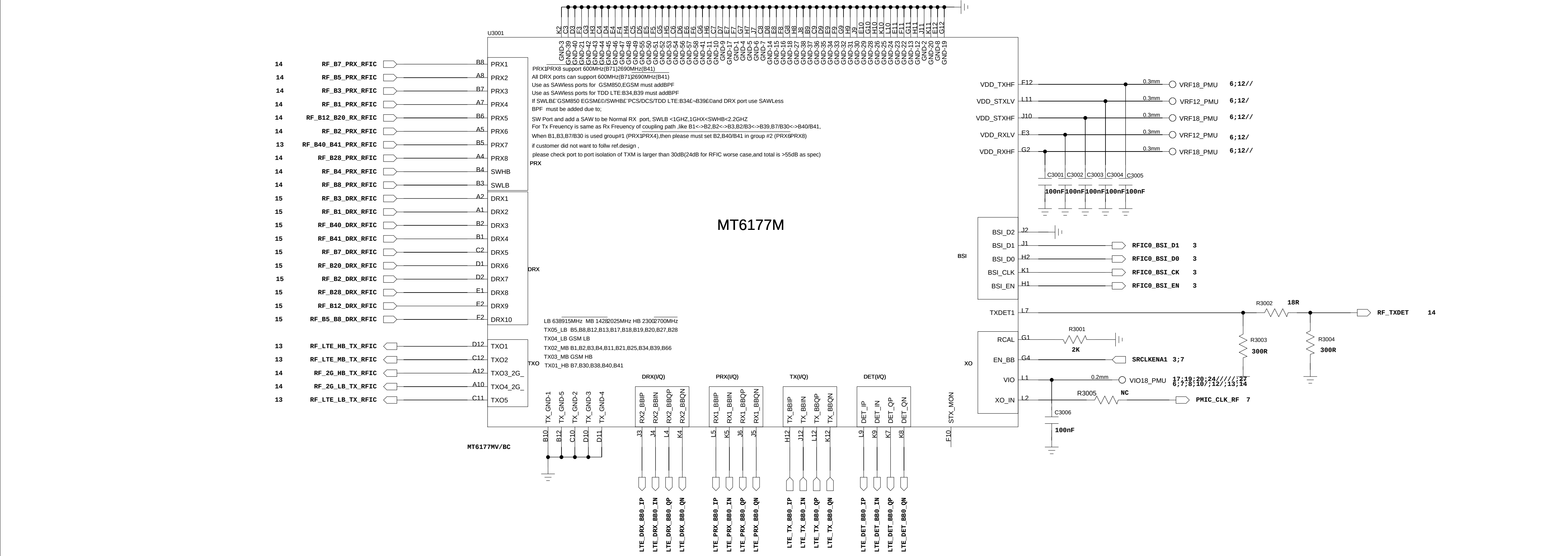
WIRELESS CHG

REVISION RECORD			
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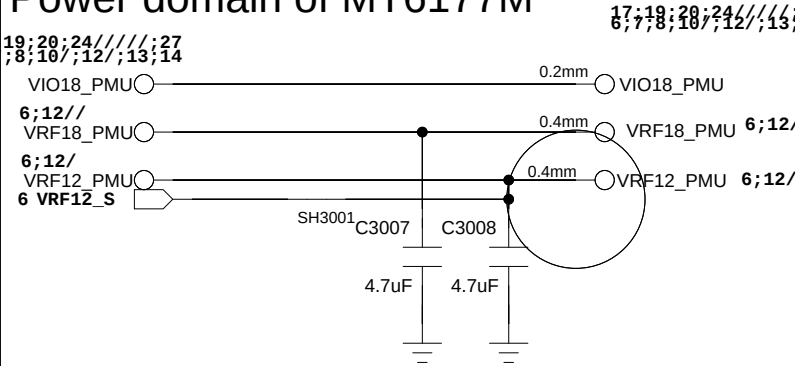
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CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

RF_MT6177M_PIN_OUT

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



Power domain of MT6177M

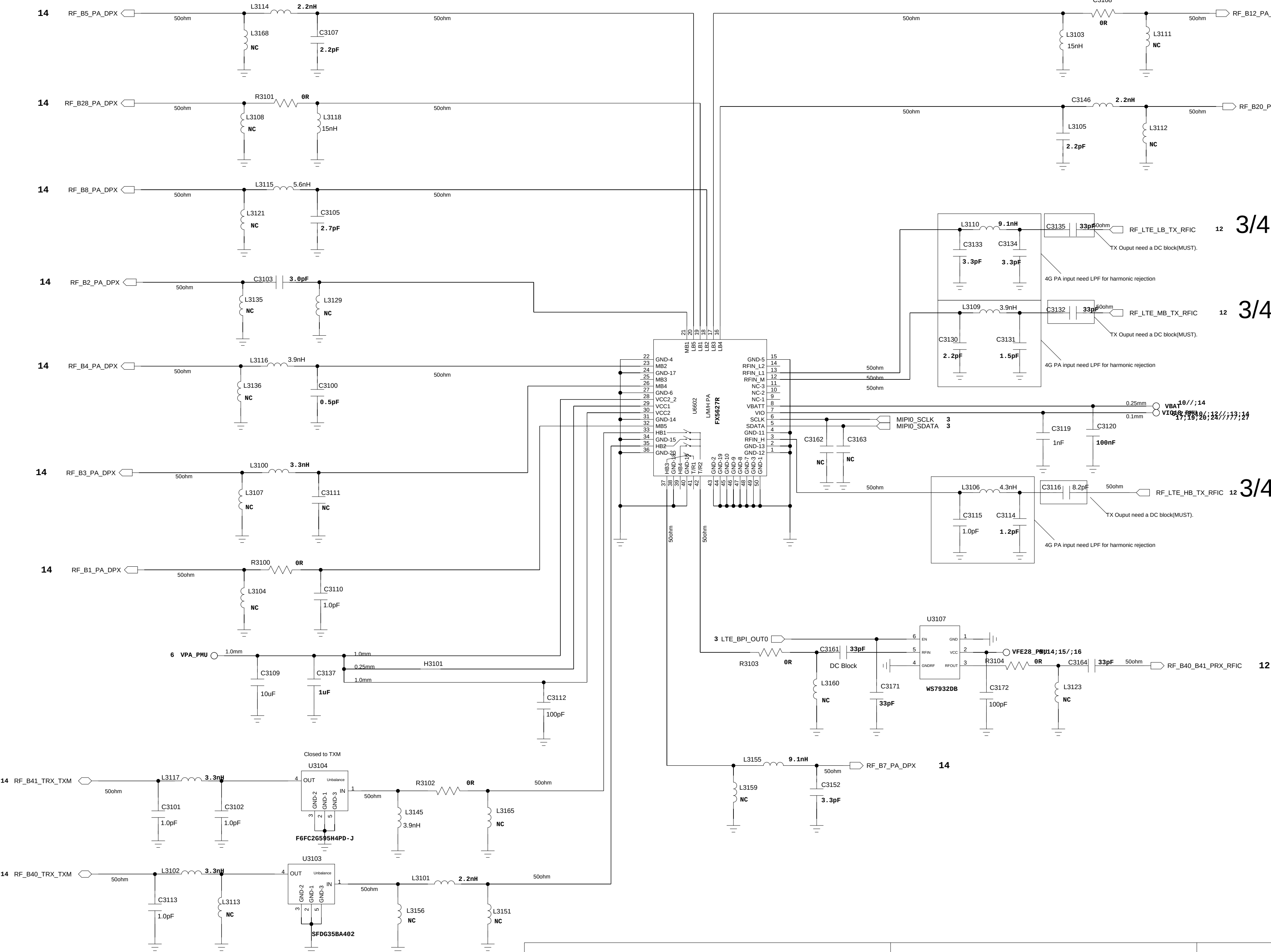


Star Power

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CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

RF_MT6177M_RF_TX

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

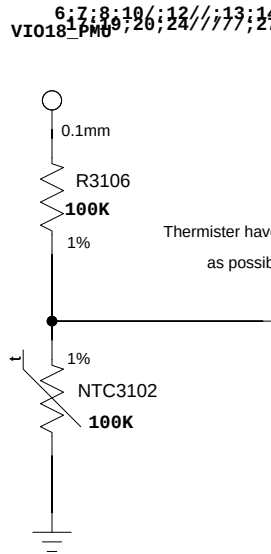


3/4G_PAIN_LB

3/4G_PAIN_MB

3/4G_PAIN_HB

Power

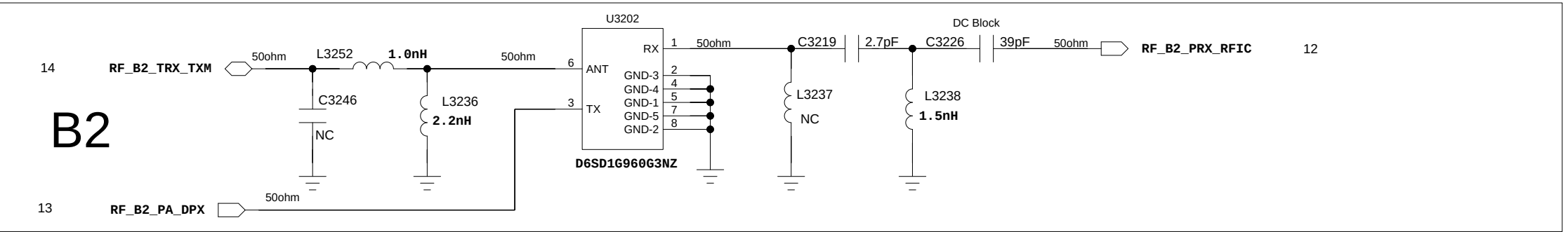
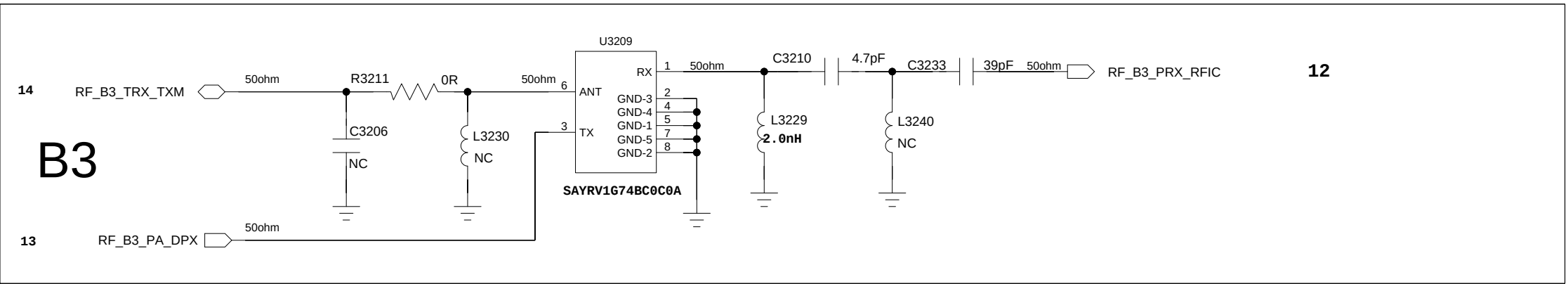
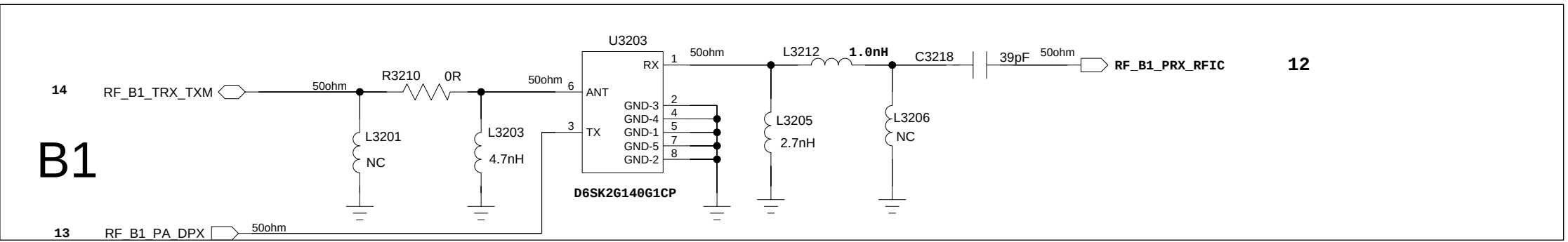
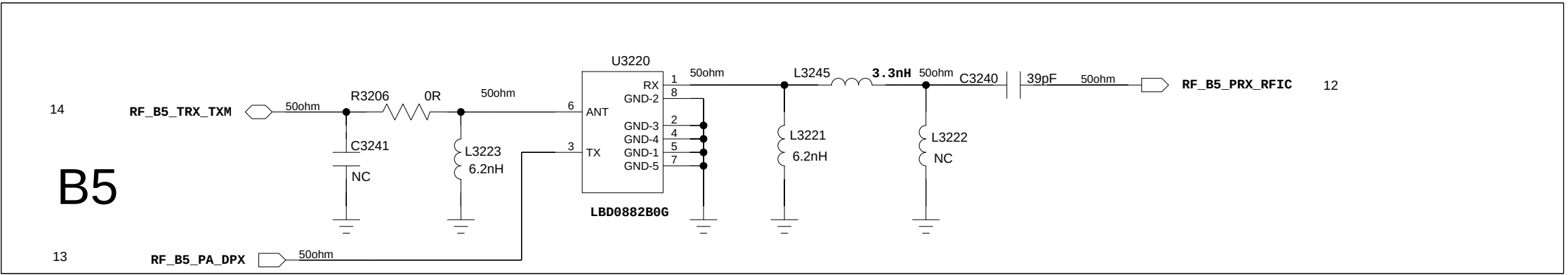
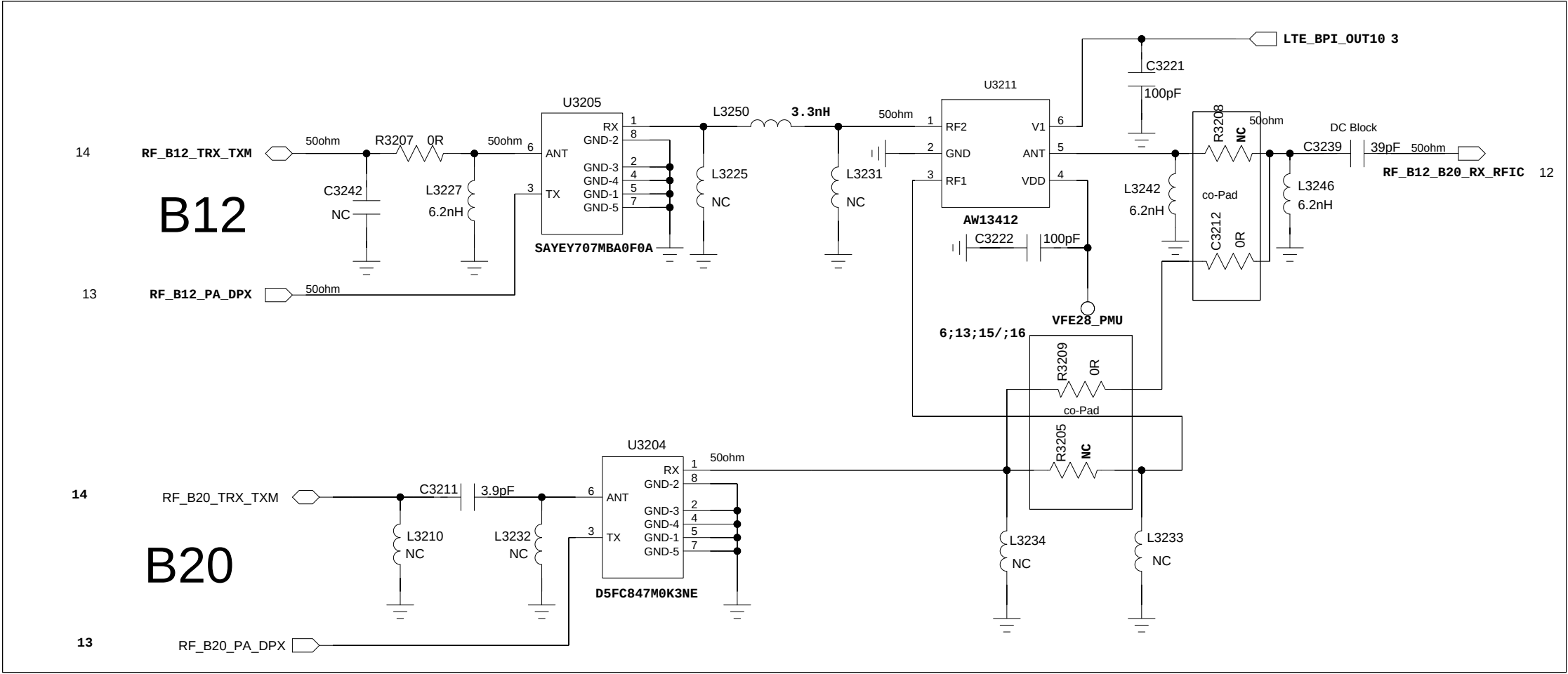
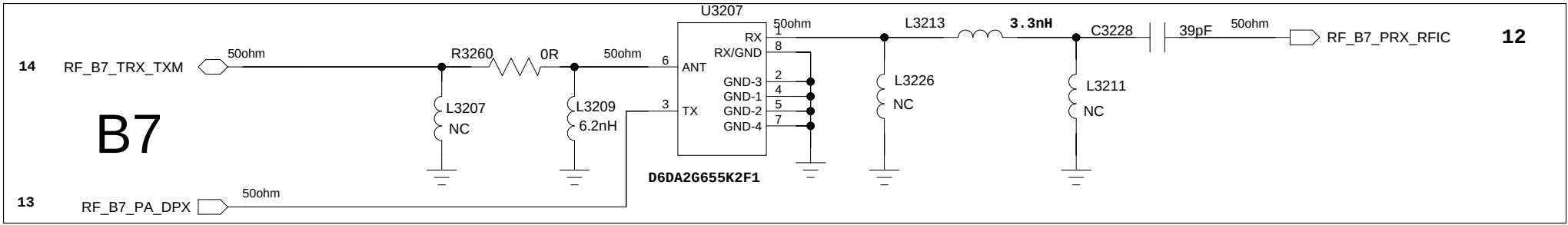
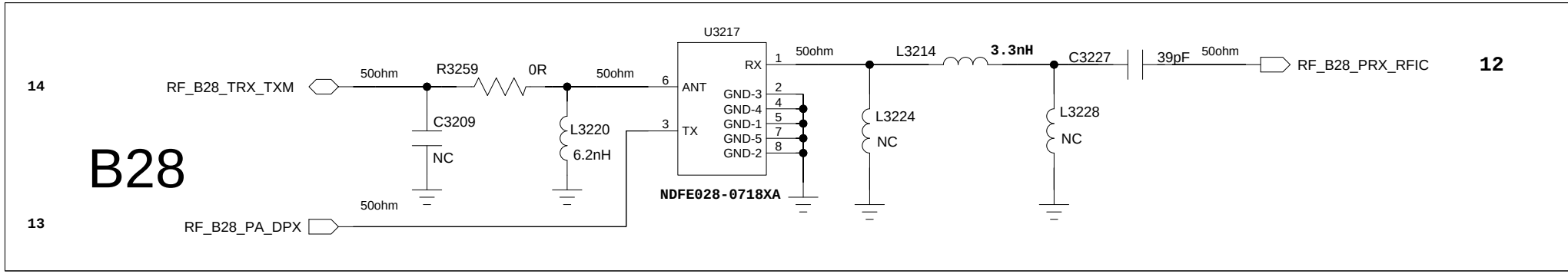
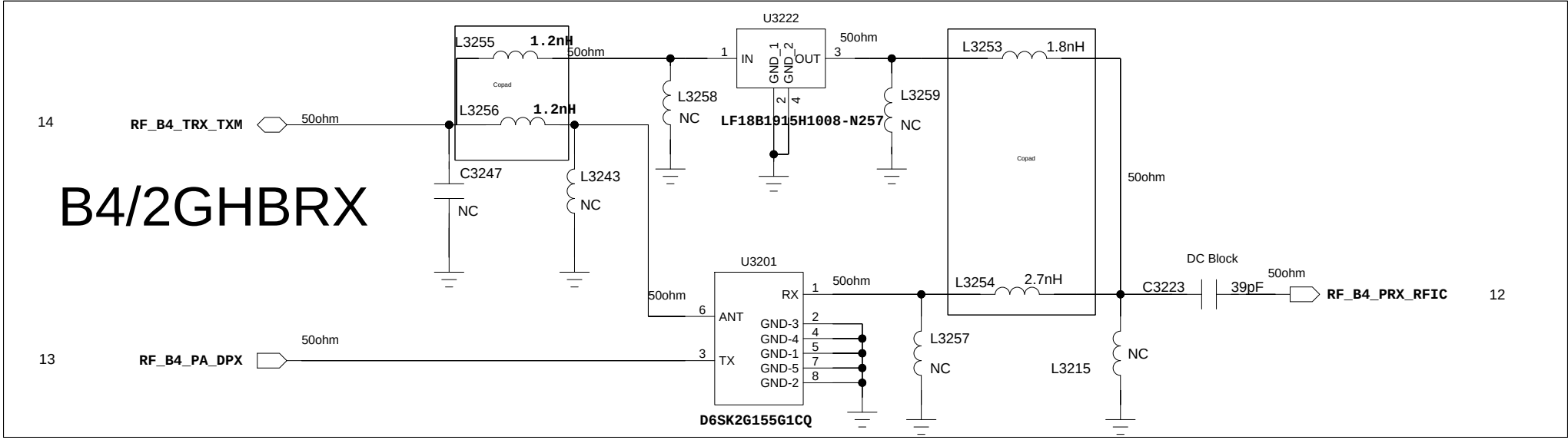
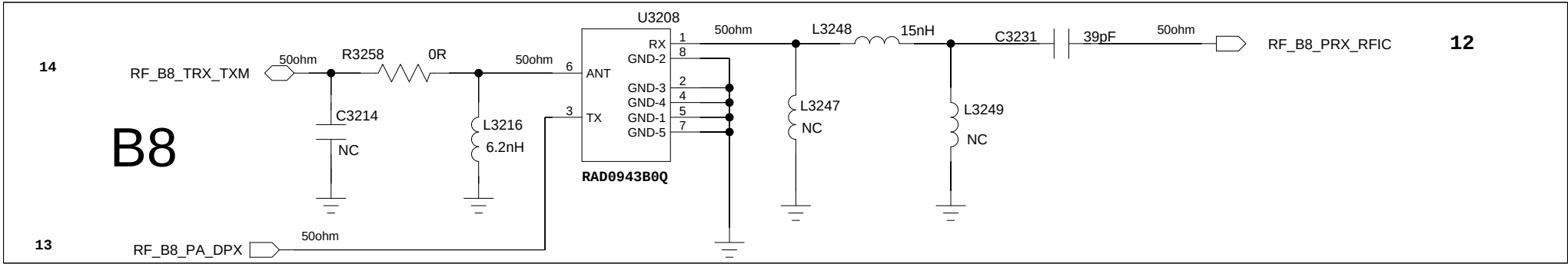
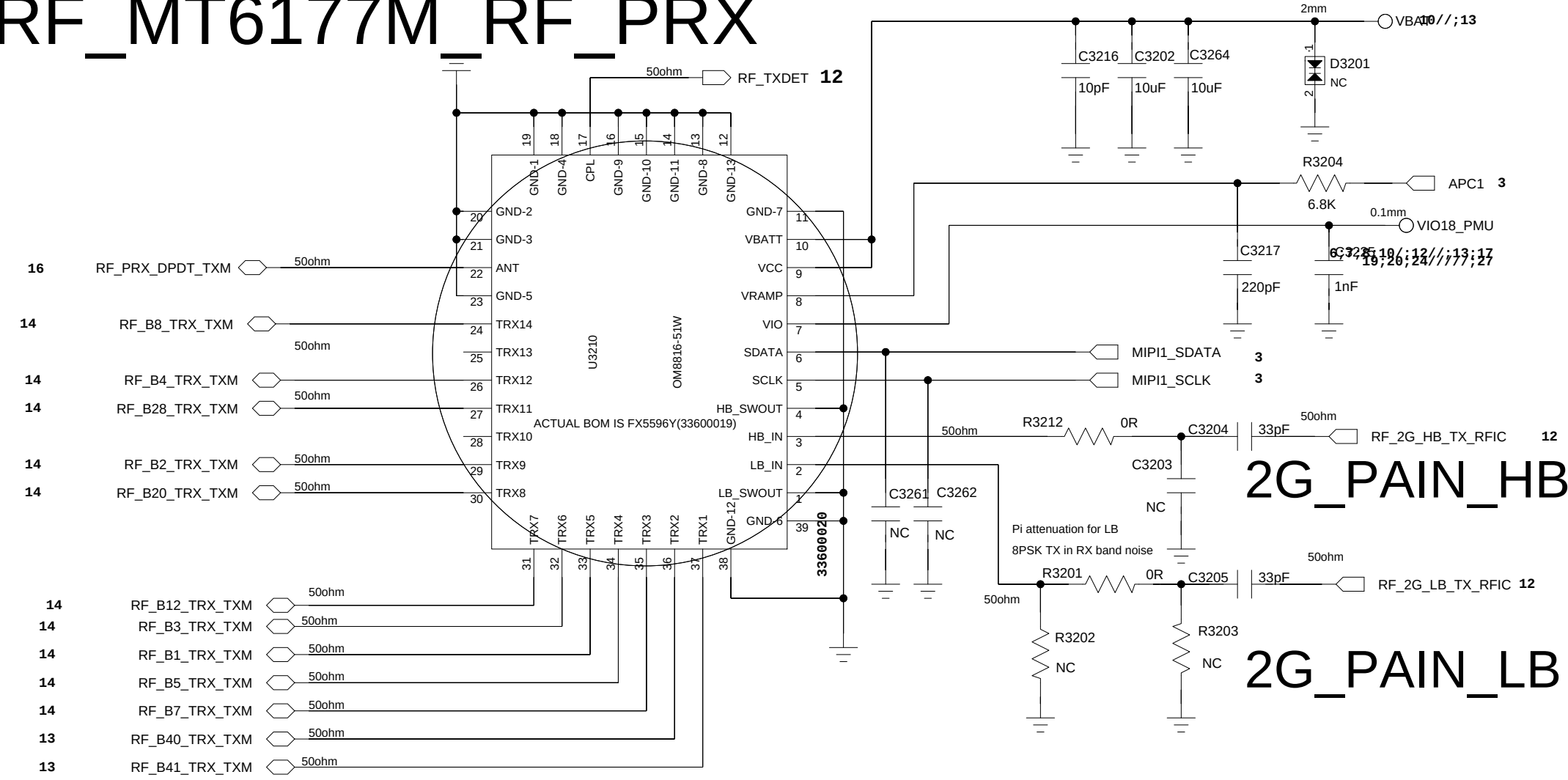


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swap B5&8 @MMMB PA 20220119

RF_MT6177M_RF_PRX

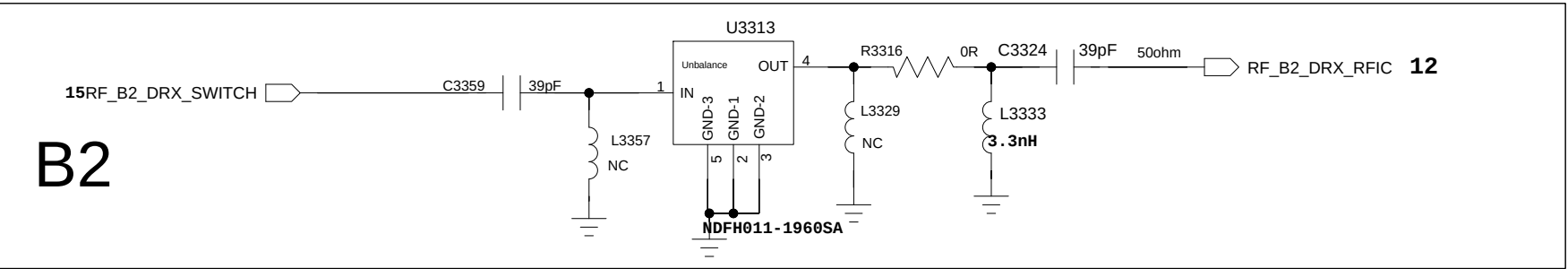
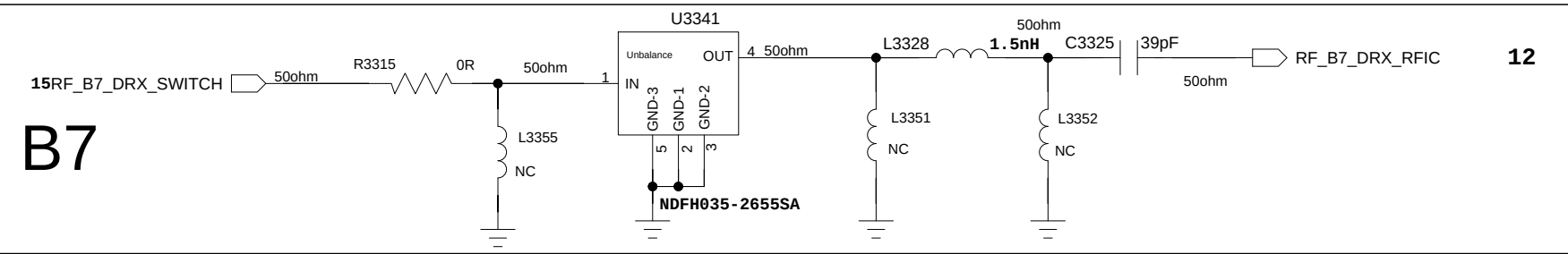
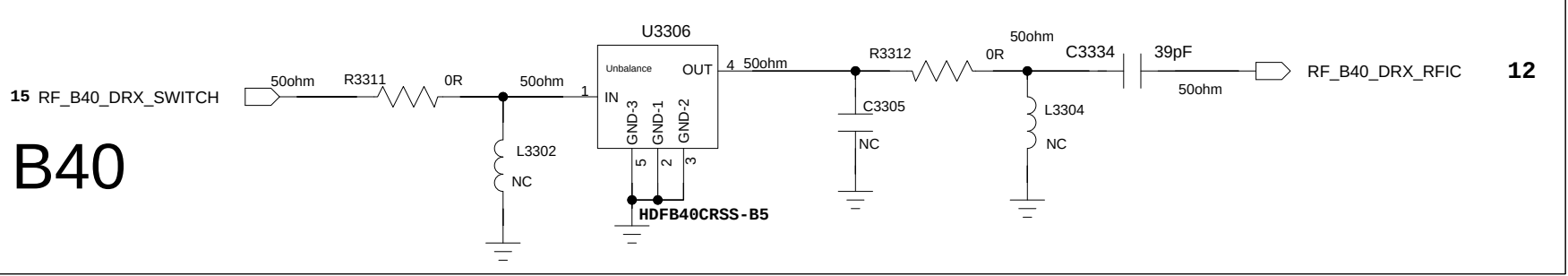
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SP8T

0.1mm



VC1613 control logic					SP8T control logic									
VC1	VC2	RF1	RF2	RF3	SP6T control logic									
V1	V2	V3	RF1	RF2	RF3	RF4	RF5	RF6	RF7	RF8				
H	H	N	Y	N	L	L	L	Y	N	N	N	N		
N	H	N	N	Y	L	L	H	N	Y	N	N	N		
					L	H	L	N	N	Y	N	N		
					L	H	H	N	N	N	Y	N		
					H	L	L	N	N	N	Y	N		
					H	L	H	N	N	N	N	Y		
					H	H	L	N	N	N	N	Y		
					H	H	H	N	N	N	N	Y		

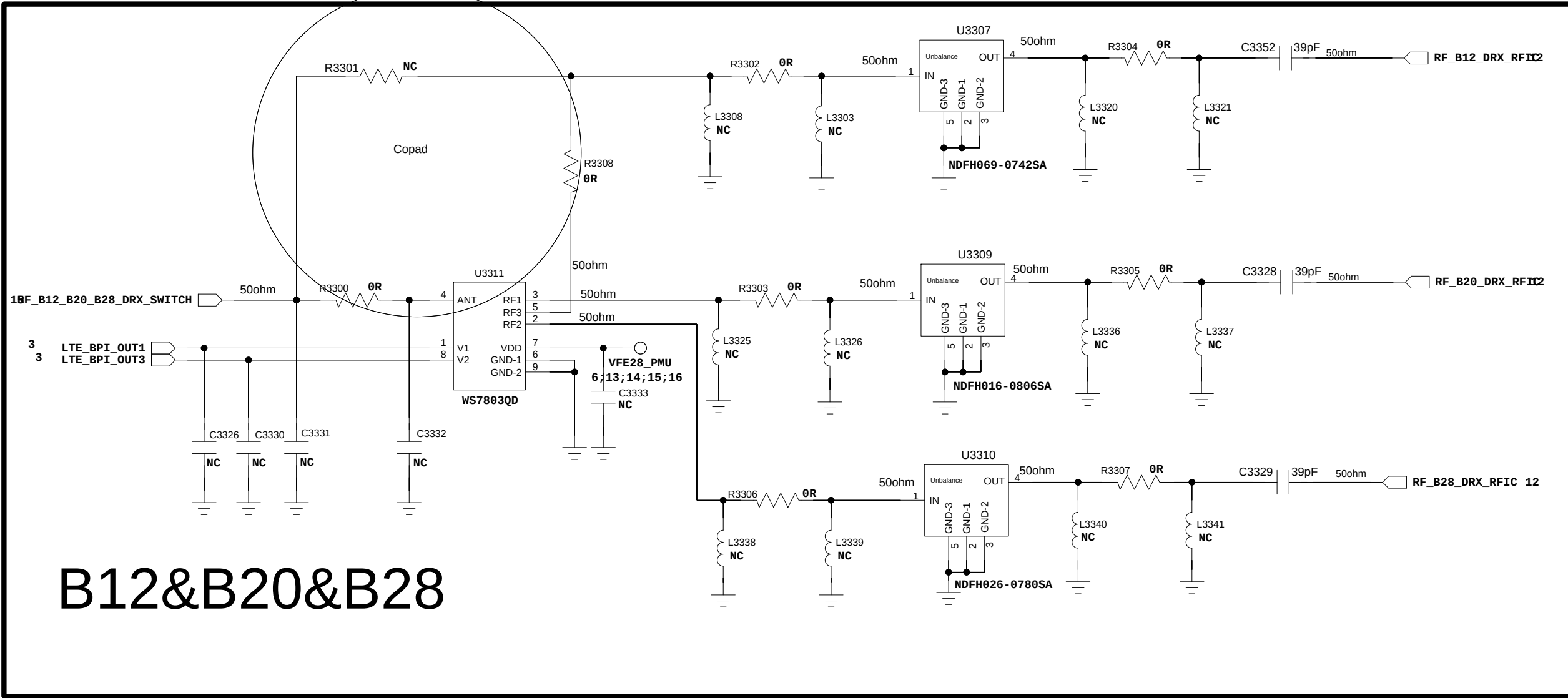
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15 **VFE28_PMU_DIV** 
R3350

0R
0.2mm
 **VFE28_PMU**

6;13;14;15;16

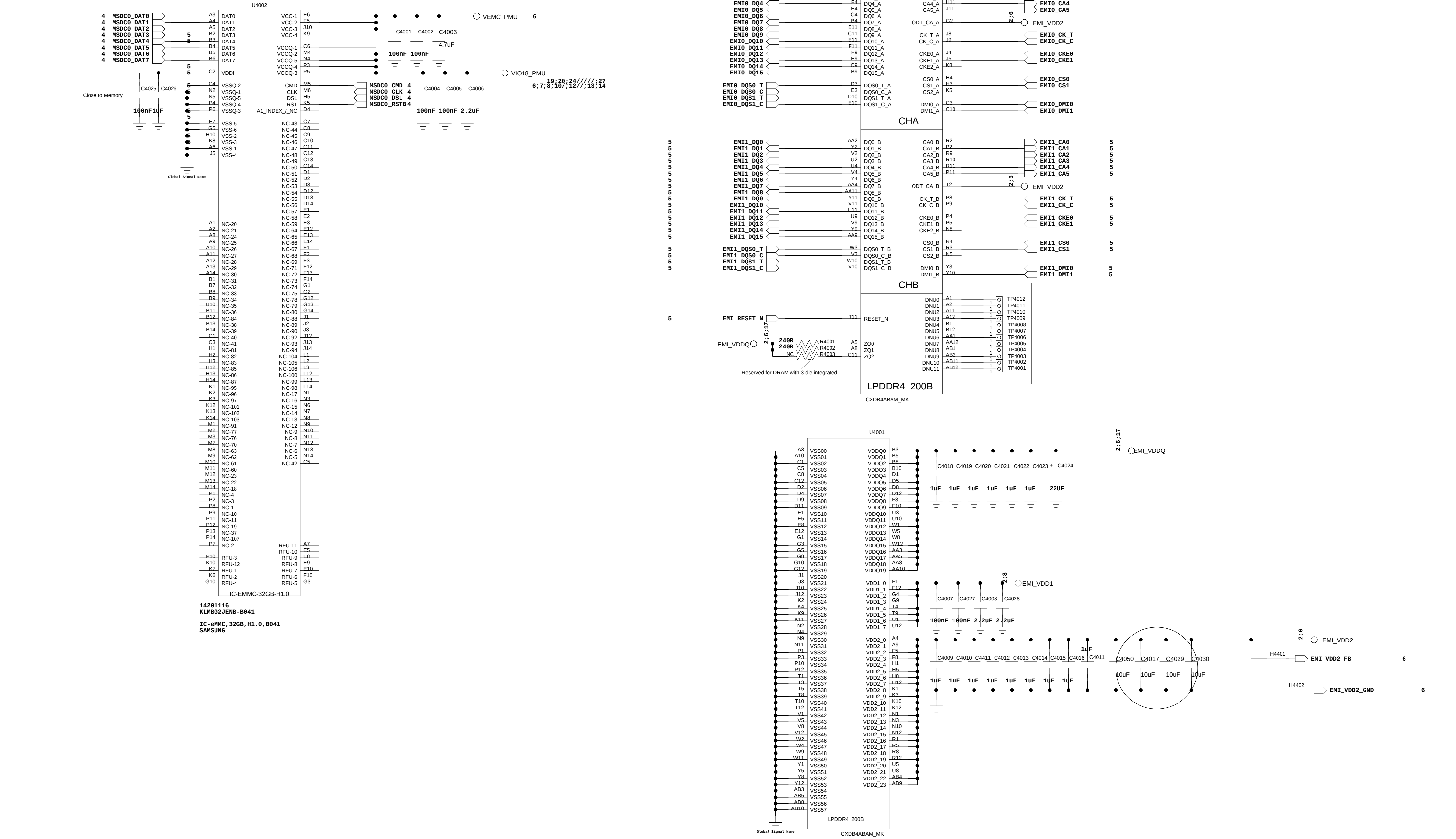
Power



COMPANY: TRANSSION HOLDINGS				MODEL: G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
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MEMORY_eMCP_LPDDR4X

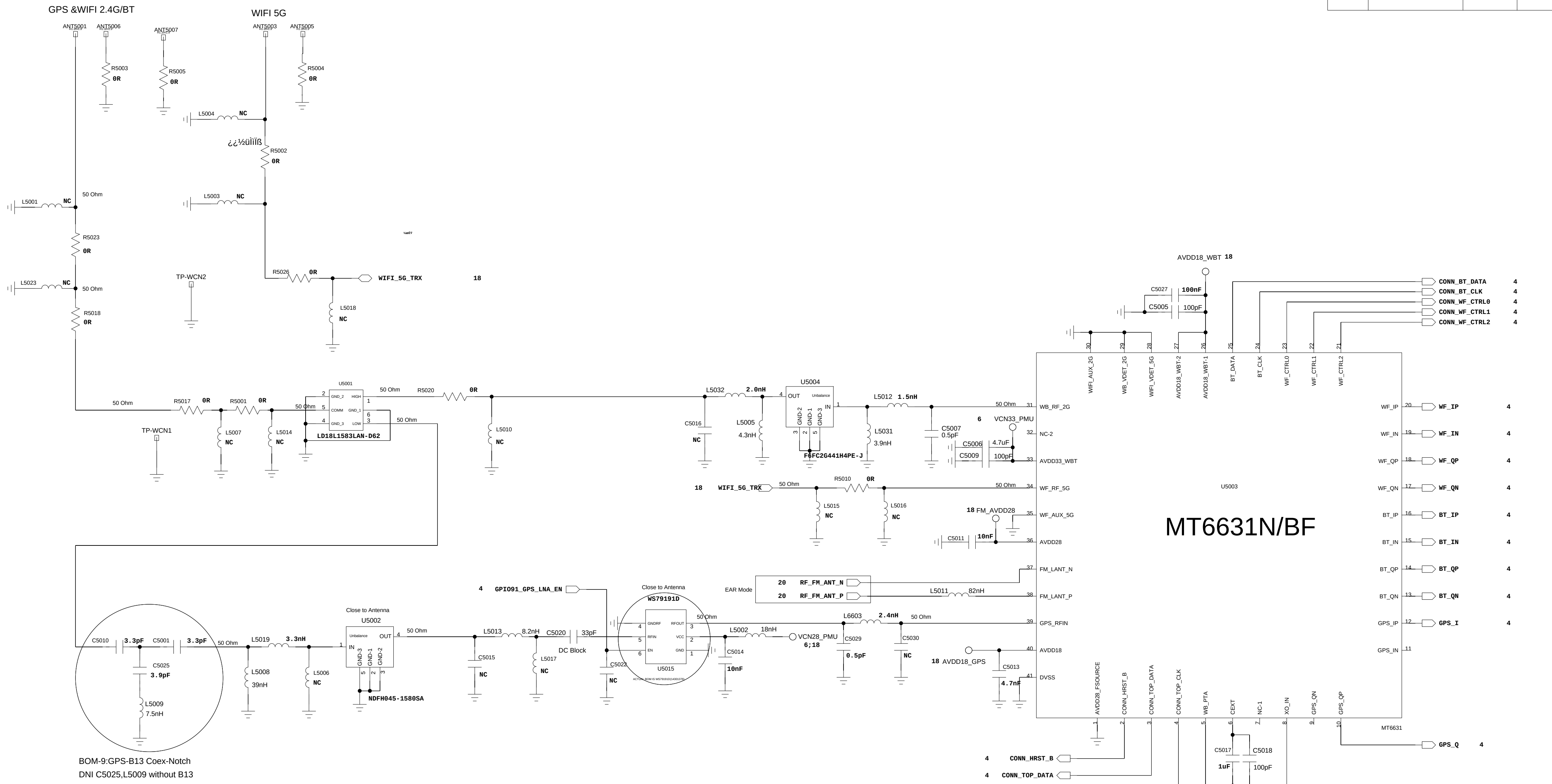
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CONNECTIVITY_MT6631

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



MT6631N/BF

Power domain

The power domain schematic shows the connections for the FM_AVDD28, AVDD18_GPS, and AVDD18_WBT power planes. It includes components like capacitors (C5019, C5012), inductors (L5003, L5004), and connectors (H5003, H5004). The schematic also shows the connection to the VCN28_PMU and VCN18_PMU.

Note: For MT6631 need to configure the VCN33_2_PMU to 2.8V by PMIC SW.

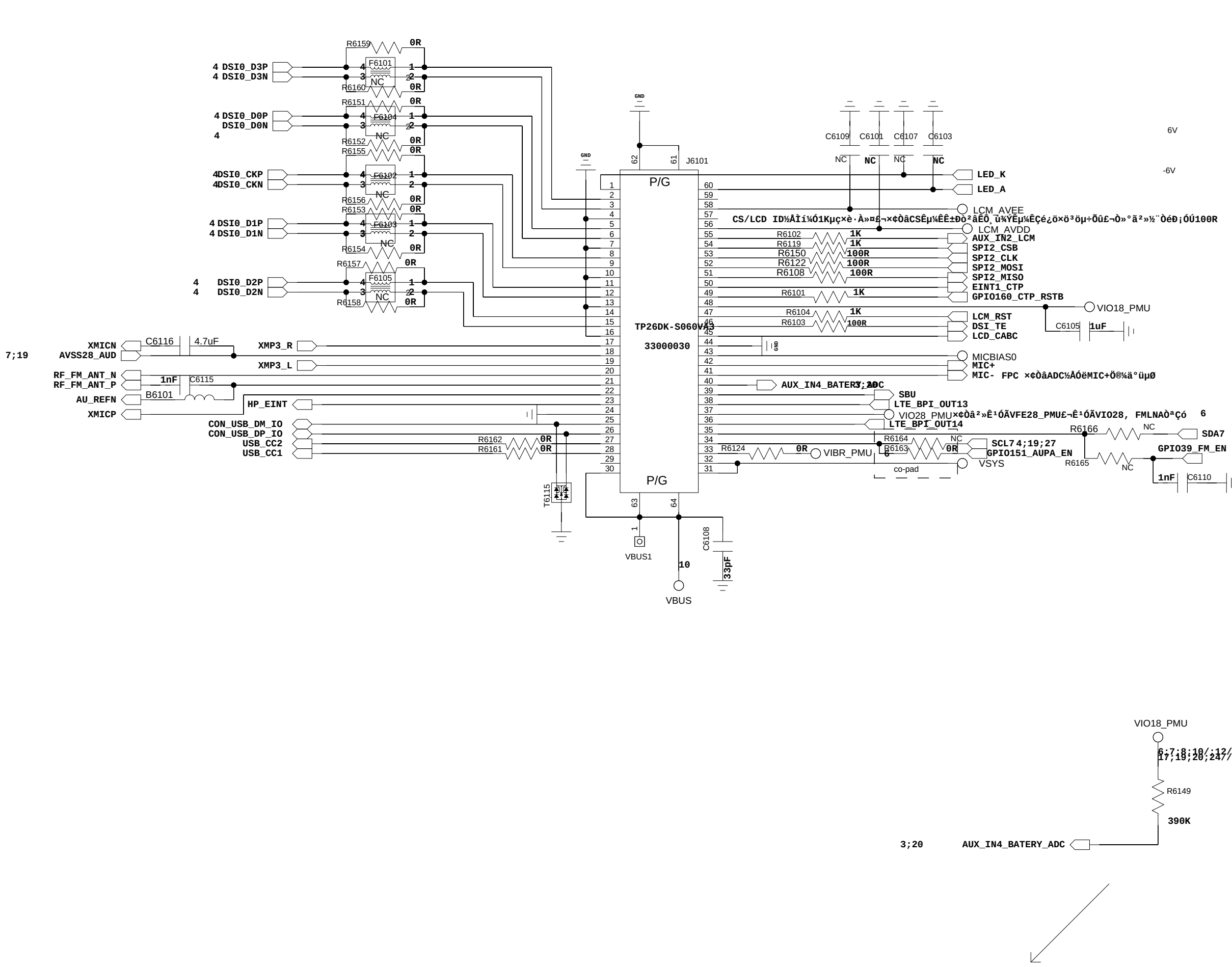
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PERI_LCM_CTP_FP

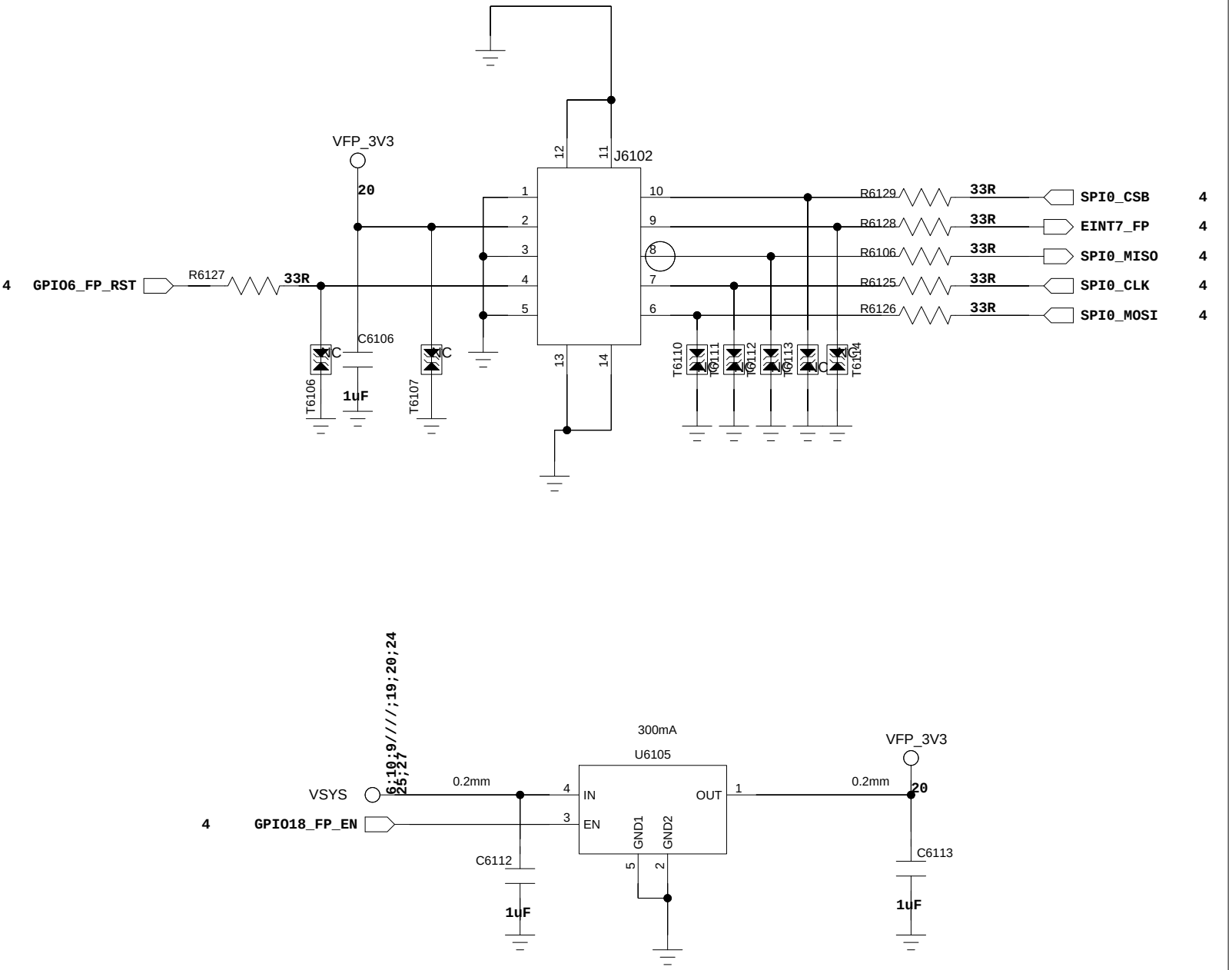
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LCD_CTP_INTERFACE

CTP (NT36525) I2C address: 0X01 (Write:0x02, Read:0x03)



FINGER PRINT



COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 12/06/2023:09:34	
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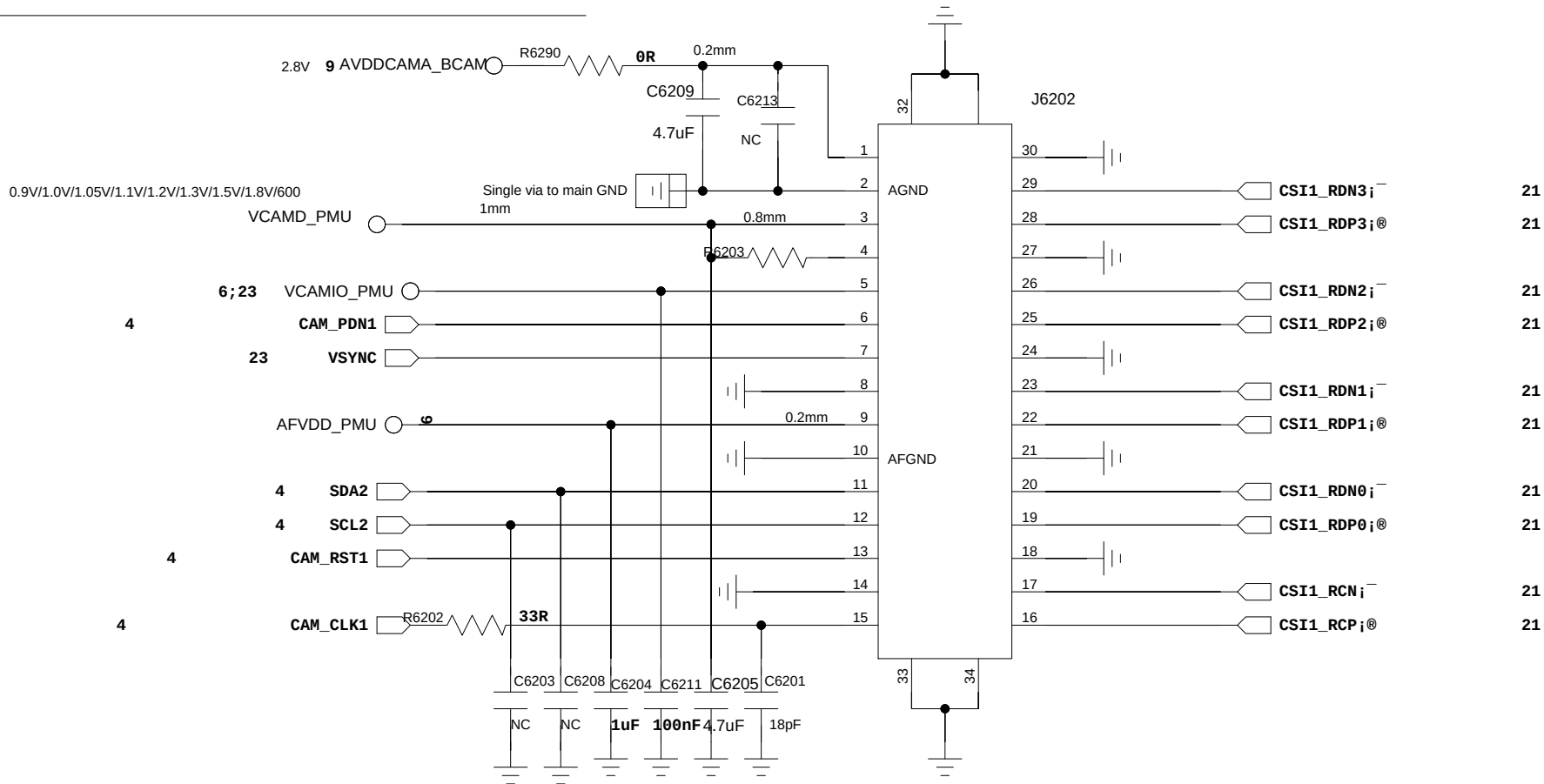
PERI_CAMERA_I

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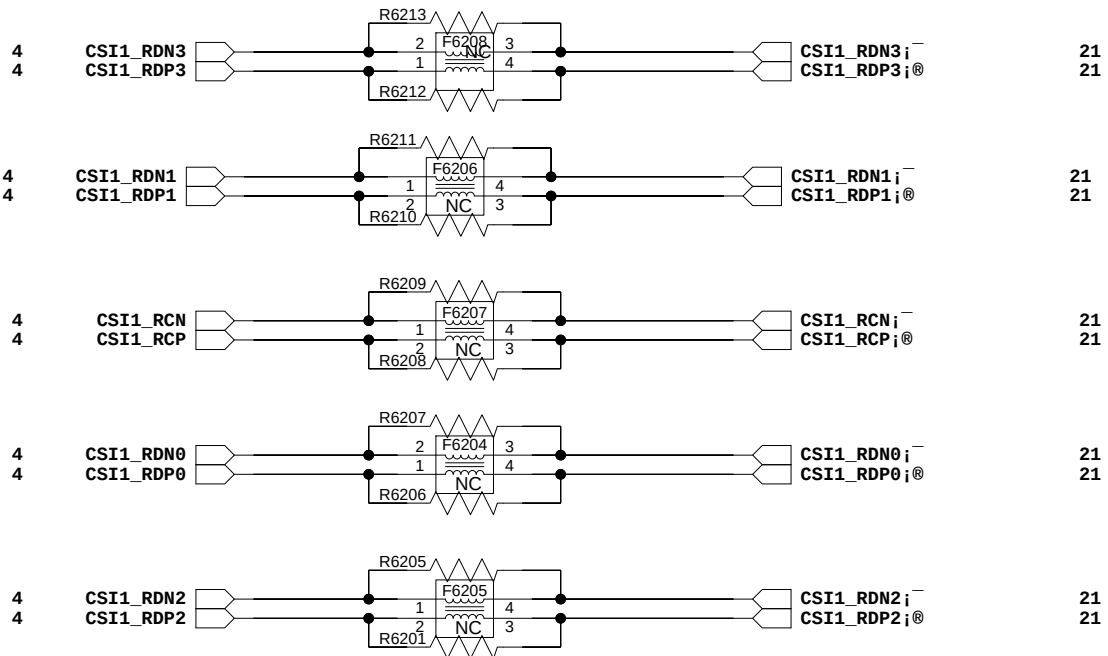
REAR CAM I (50M/13M/8M)

64M/50M/48M R6203=0R VCAMD_EXT1¹@μç

13M/16M/8M R6203=NC



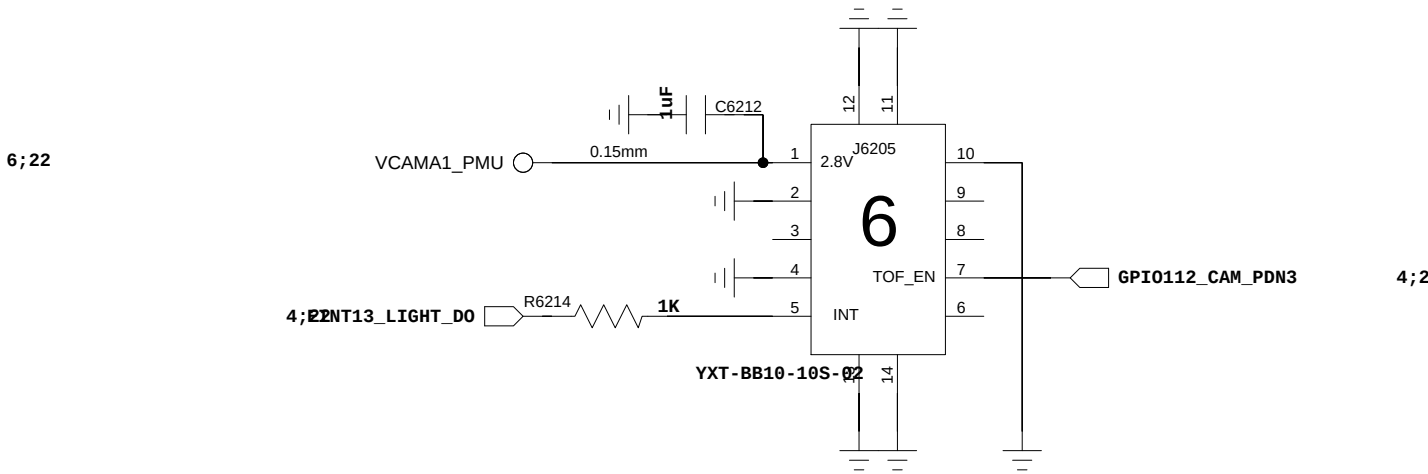
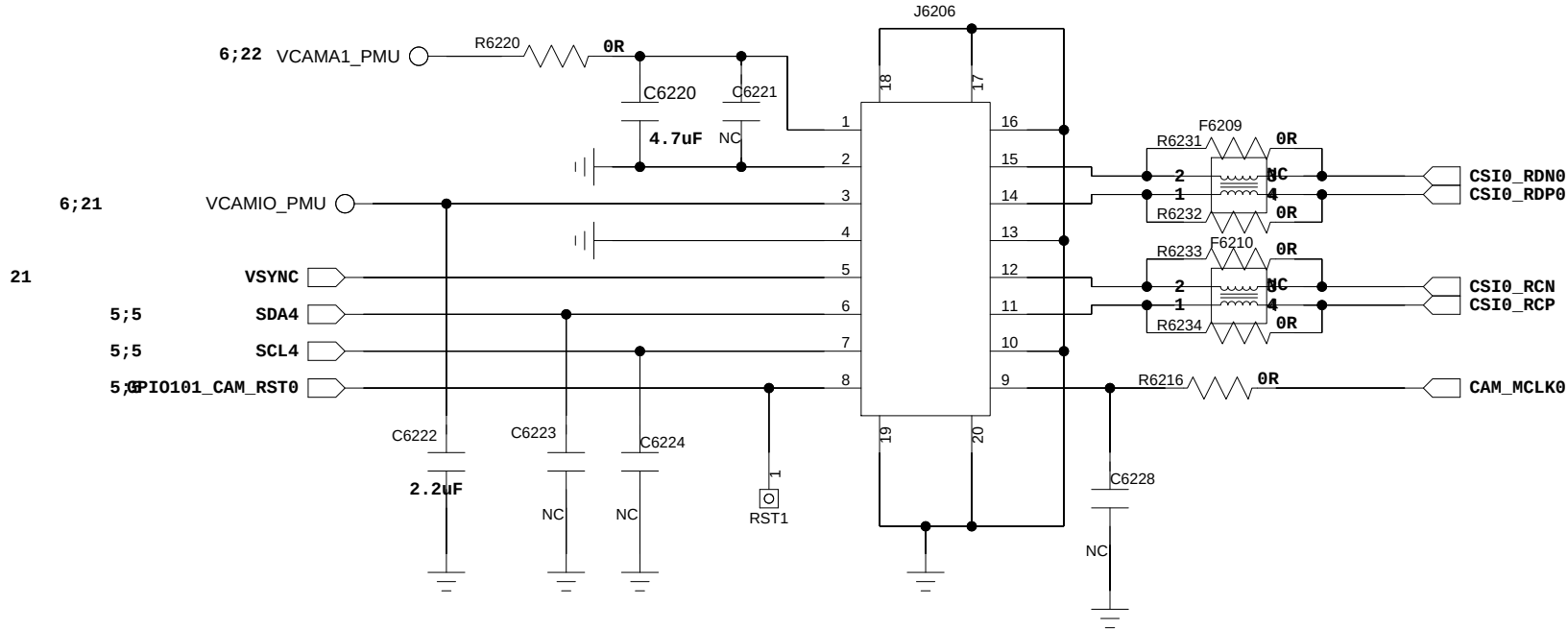
	SENSOR	I2C Address	VCAMA	VCAMD	VCAMIO	AFVDD	SAP
50M	Hi-5021Q AA5021PXX	Write:0x40 Read:0x41	2.8V @78mA	1.1V @598mA	1.8V@ 31mA	2.8V@ 103mA	17201826(0.7um)
	SSKJN1S03-FGX9	Write:0x20 Read:0x21	2.8V @67.7mA	1.05V @260.7mA	1.8V@ 4mA	2.8V@ 100mA	17201711(0.64um)
13M	Hi-1336-AA1336NXX-C20AHX	Write:0x40 Read:0x41	2.8V @44mA	1.1V @106mA	1.8V@ 6mA	2.8V@ 125mA	17201835
	SSK3L6X03-FGX	Write:0x20 Read:0x21	2.8V @45mA	1.05V @160mA	1.8V@ 1.1mA	2.8V@ 125mA	17201836
8M	Hi846/AA08466NXX-B2CWCX	Write:0x40 Read:0x41	2.8V @42mA	1.2V @67mA	1.8V@ 2mA	2.8V@ 123mA	17201833
	GC08A3-WA1XA	Write:0x62 Read:0x63	2.8V @60mA	1.2V @150mA	1.8V@ 10mA	2.8V@ 125mA	17201834



COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 62_PERI_CAMERA_I		VERSION:	SHEET: 21 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

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LTR	ECO NO:	APPROVED:	DATE:

2M



1 ãÃô

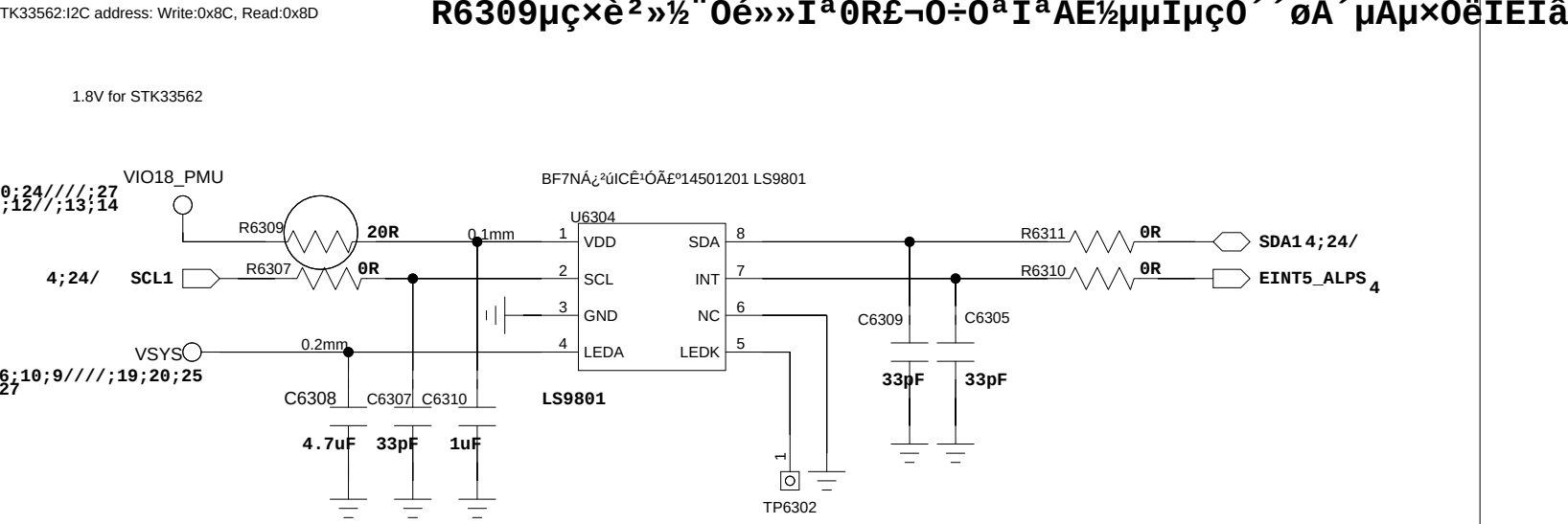
			VCAMA	PIN5			SAP
			2.8	1.7+0.2			17201729
1 ãÃô			2.8				17201744
			2.8				17201845

COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 62_PERI_CAMERA_II		VERSION:	SHEET: 22 OF 27
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PERI_SENSORS

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AL& PS Sensor



G-Sensor + Gyro Sensor

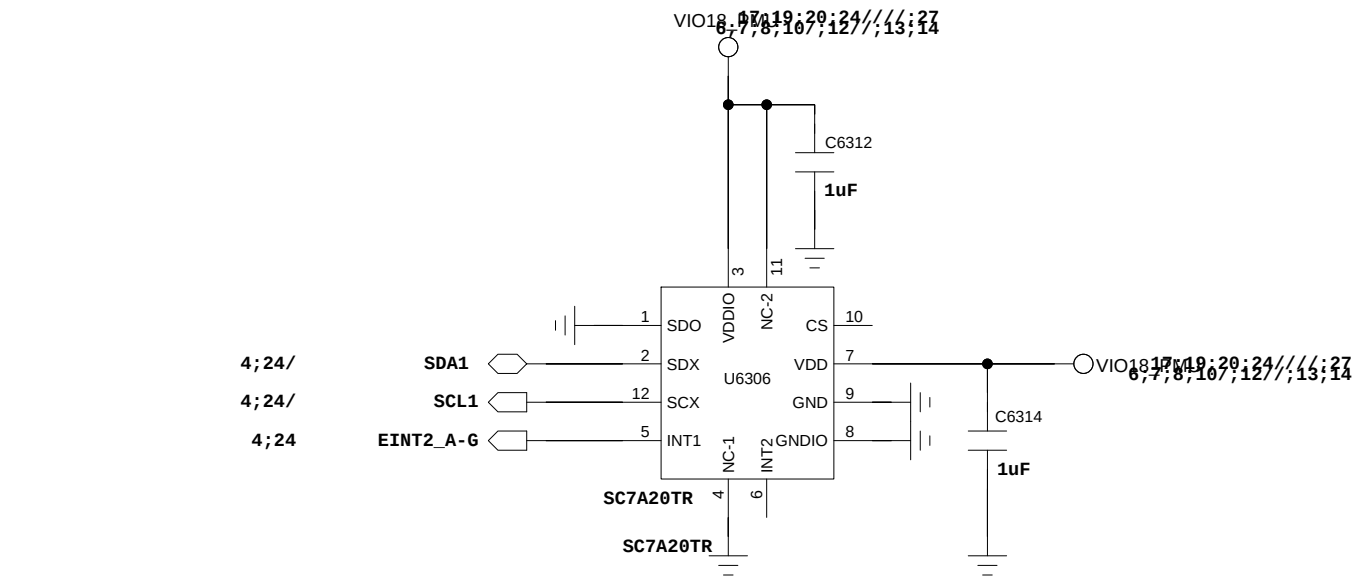
LSM6DS3TR-C:I2C address: Write:0xD4, Read:0xD5

G-Sensor

MXC6655XA:ADDRESS:0x2A(Write)/0x2B(Read)

SC7A20:ADDRESS:0x30(Write)/0x31(Read)

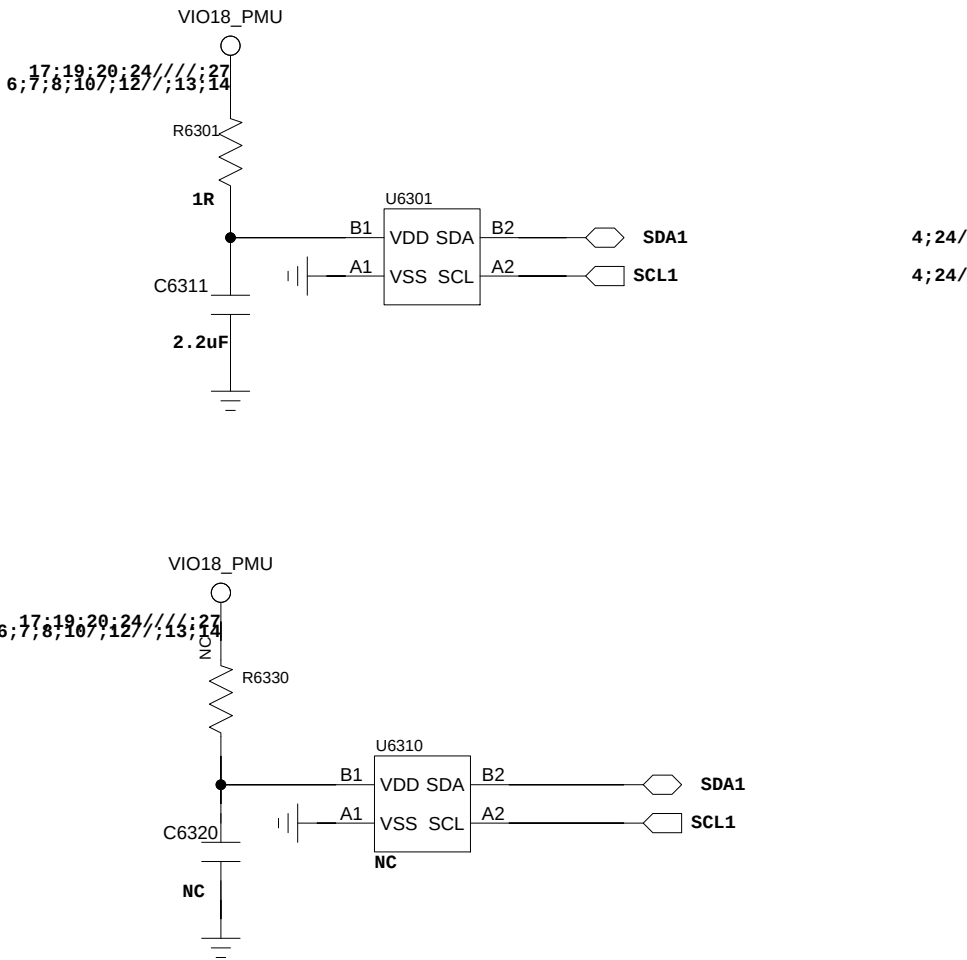
MXC4005XC:ADDRESS:0x2A(Write)/0x2B(Read)



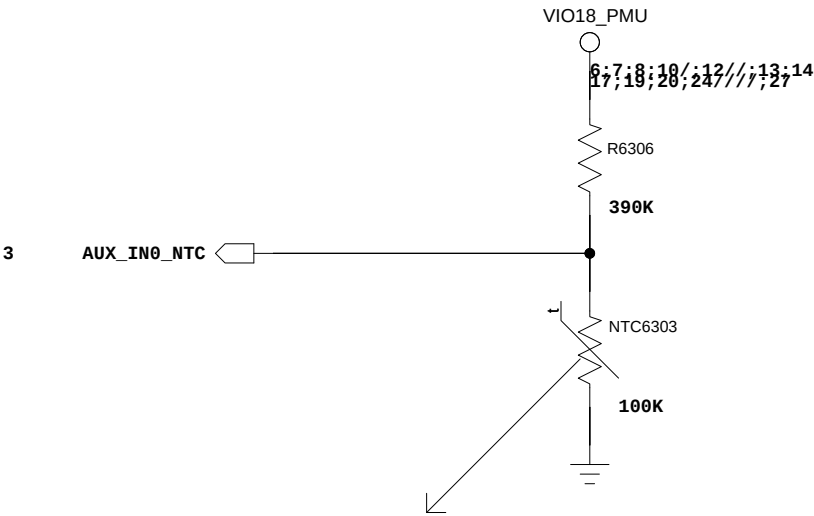
M-Sensor(COMPASS)

MMC5603:I2C ADDRESS:0x60(Write)/0x61(Read)

QMC6308:I2C ADDRESS:0x58(Write)/0x59(Read)



Thermistor to sense AP temperature



1. NTC6303must keep a distance about 68 mm away from BB and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

COMPANY: TRANSSION HOLDINGS

MODEL:G85_STD_MAIN

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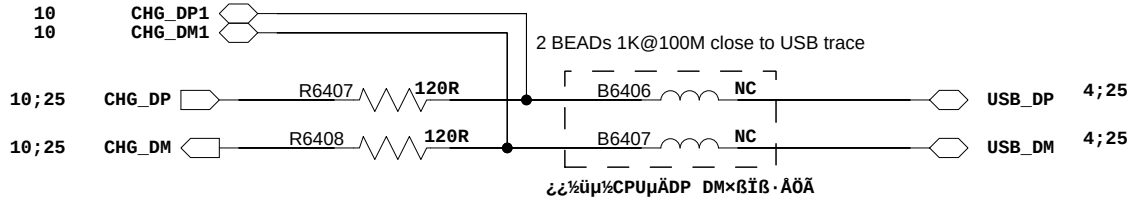
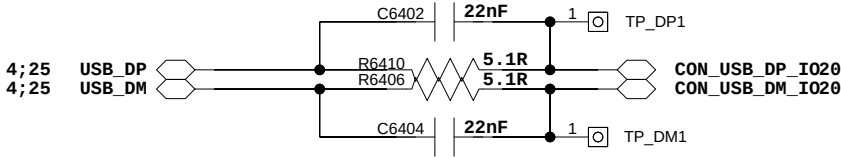
component	MXC6655XA	SC7A20	STK8BA50-S
C6302	NC	100nF	1uF
C6303	1uF	100nF	1uF

PERI_EXCH_IO

REVISION RECORD			
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RFC

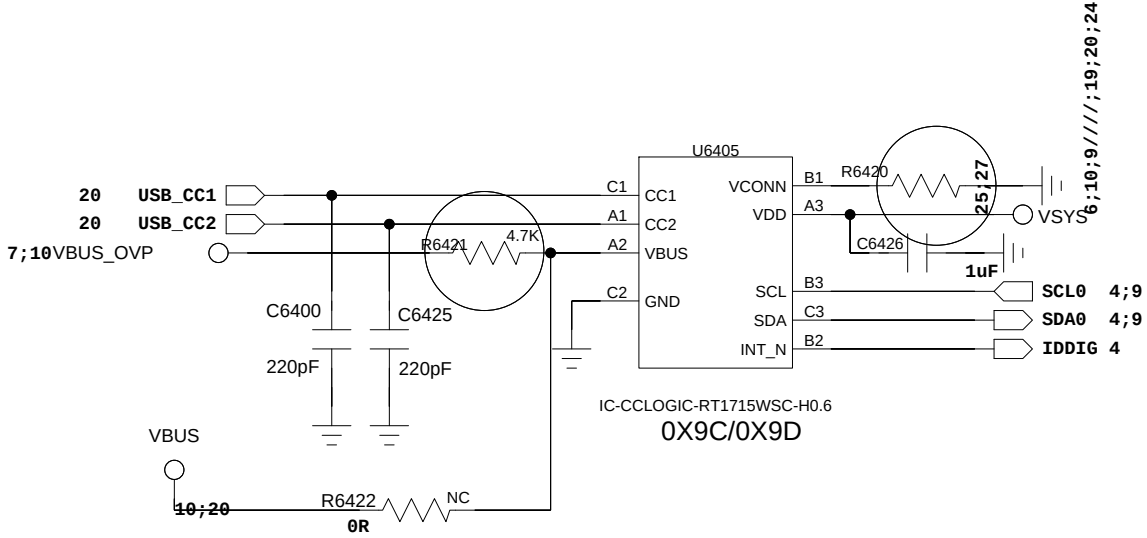
Put it on the trace,do not by-passed Place on battery side outside shielding case



Ð-ðéSMTμç×èĖμÃ÷

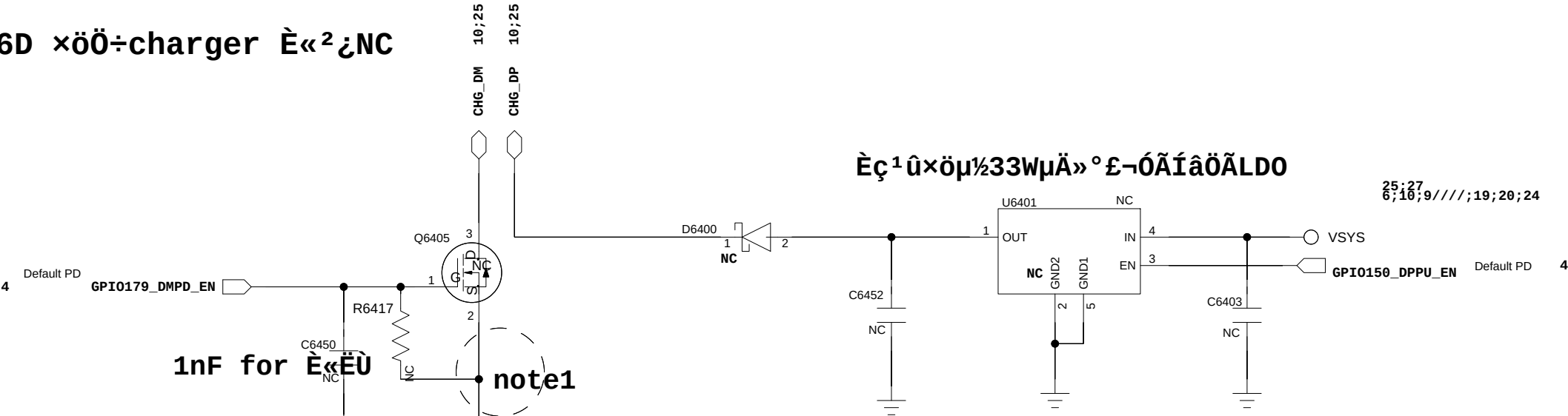
18W IC	T6405	R6407	R6408	ĖμÃ÷
OTHERS	RES-->0R	120R	120R	T6405 SMT 0R

CC Logic



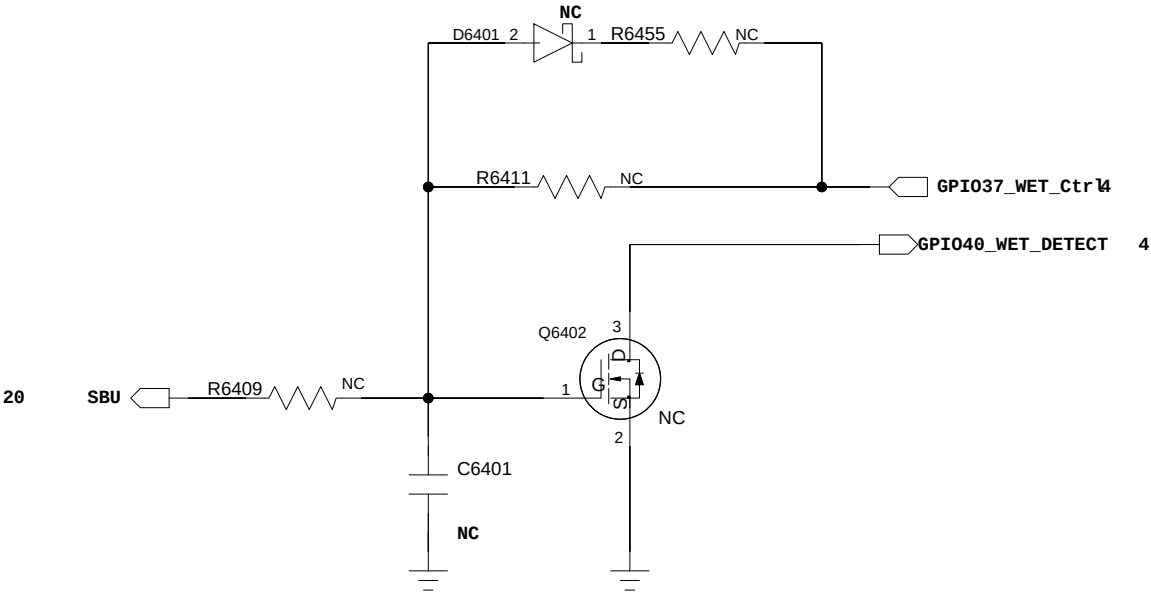
HVDCP TA

GM41516D ×öð÷charger È«²¿NC

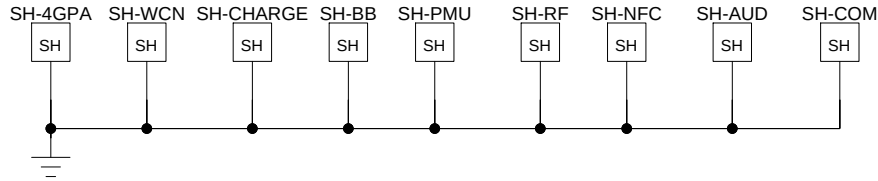


note1£°RT9467/ETA69630ÃμÄQC0ðÆ-μçÂ· ¡£Ė¹óÄRT9470D/SGM41516DμÄĖ±°ð¶p%«¹ÜNCö±½óİÂGND¡£

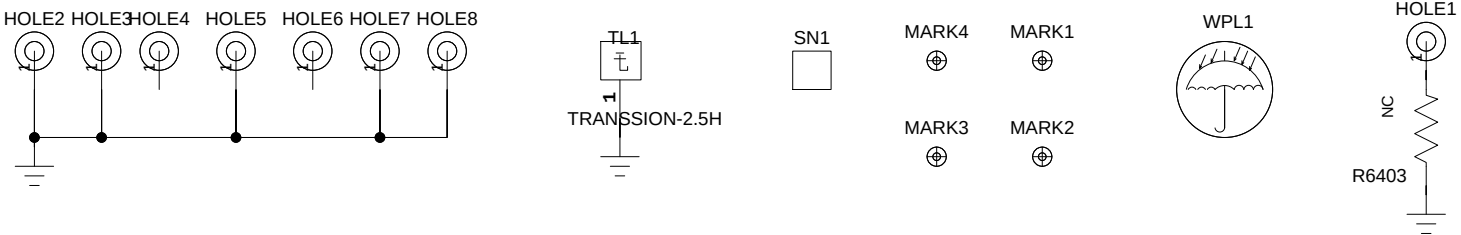
WET detect



SHIELDING



LOGO&SN&MARK

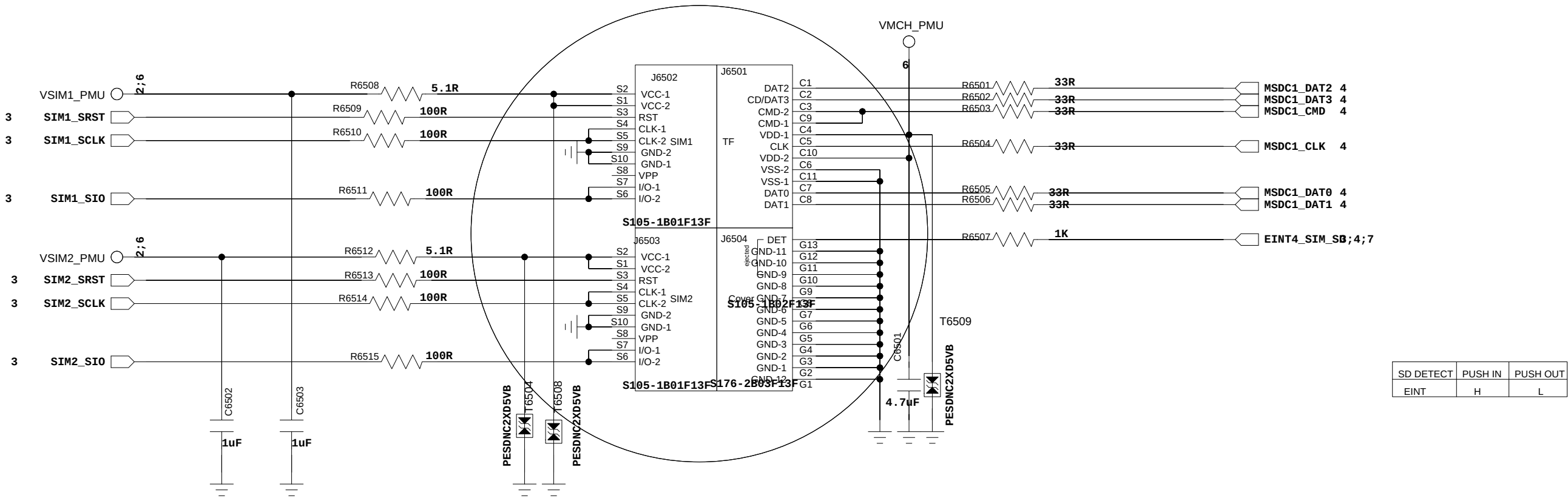


COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 13/06/2023:11:15	
DRAWN		DATED		TITLE: 64_PERI_EXCH_IO		VERSION:	SHEET: 25 OF 27
CHECKED		DATED	< >	Confidentiality	CONFIDENTIAL		

PERI_SIM_SD_KEYPAD

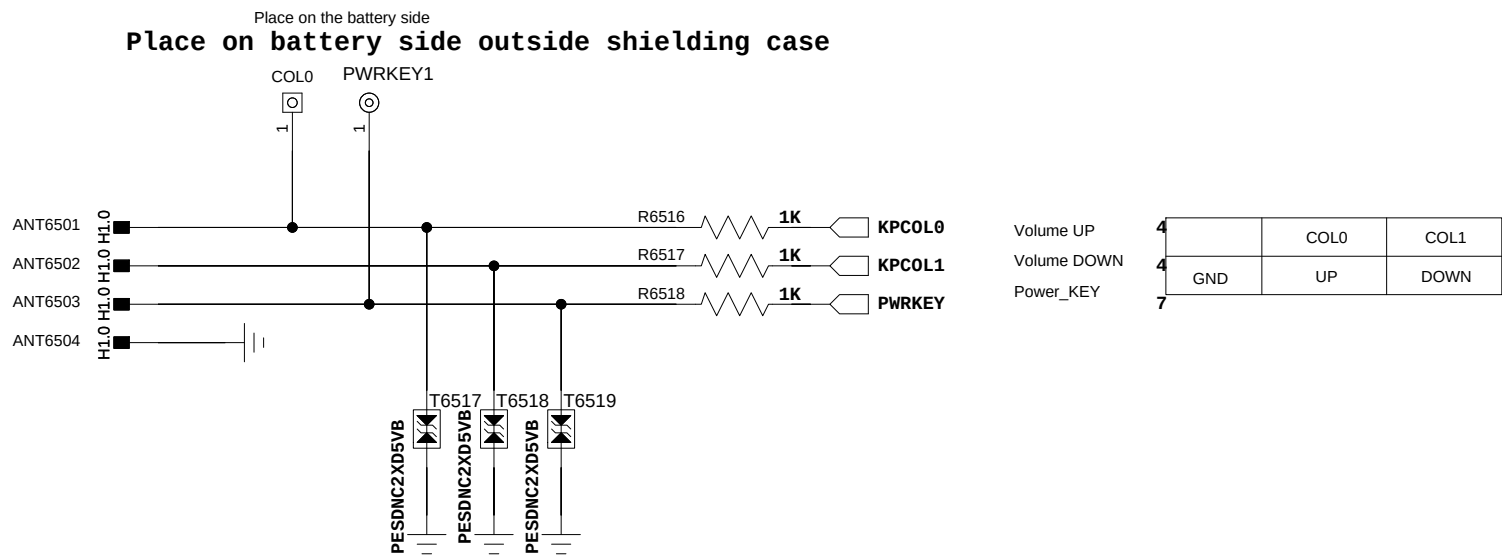
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SIM1/2-SD-CARD



SD POWER CONTROL

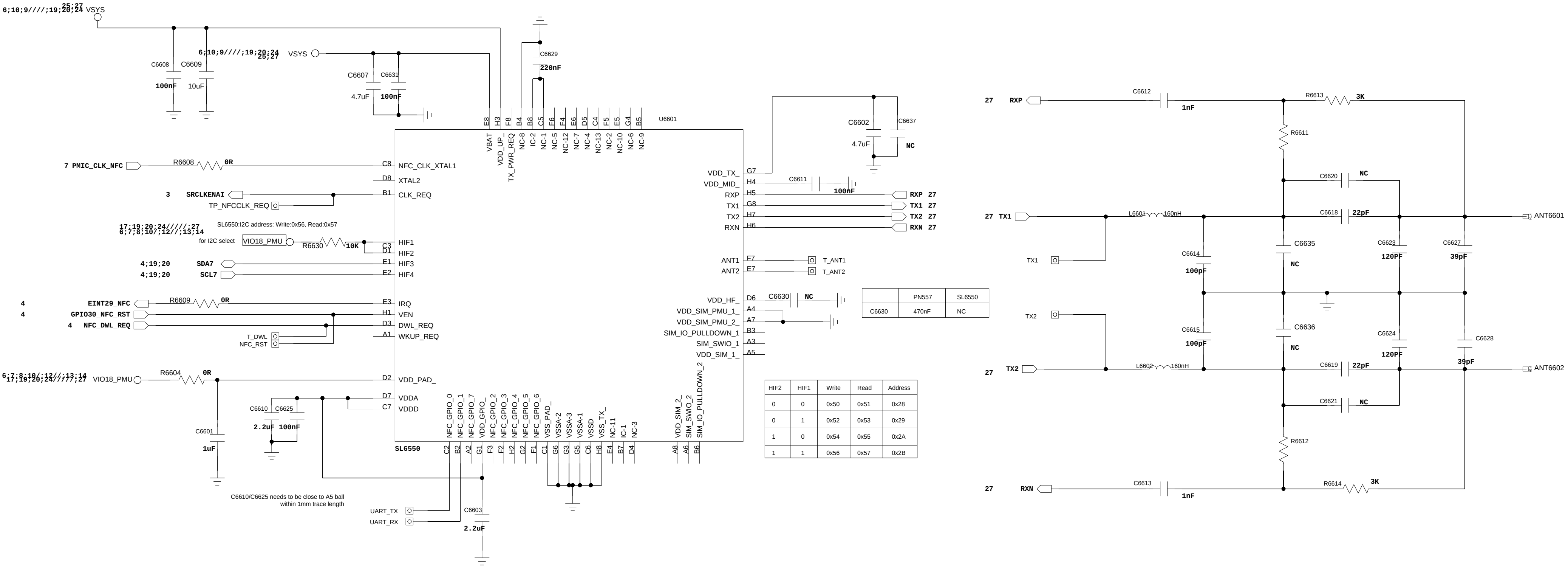
SIDEKEY



COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 65_PERI_SIM_SD_KEYPAD		VERSION:	SHEET: 26 OF 27
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PERI_NFC

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COMPANY: TRANSSION HOLDINGS				MODEL:G85_STD_MAIN		Modified Date: 10/06/2023:19:27	
DRAWN		DATED		TITLE: 66_PERI_NFC		VERSION:	SHEET: 27 OF 27
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